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Hawkinson

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(54) **INTEGRATED GUIDE SYSTEM AND DOOR SEAL FOR A SOFT CLOSE SLIDING DOOR**

(71) Applicant: **ASSA ABLOY Accessories and Door Controls Group, Inc.**, Monroe, NC (US)

(72) Inventor: **William Hawkinson**, Germantown, TN (US)

(73) Assignee: **ASSA ABLOY Accessories and Door Controls Group, Inc.**, Monroe, NC (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 1216 days.

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Related U.S. Application Data

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(60) Provisional application No. 62/654,662, filed on Apr. 9, 2018.

(51) **Int. Cl.**

E06B 7/23 (2006.01)
E05D 15/06 (2006.01)
E06B 7/215 (2006.01)

(52) **U.S. Cl.**

CPC ***E06B 7/2312*** (2013.01); *E05D 15/063* (2013.01); *E05Y 2201/64* (2013.01); *E05Y 2900/132* (2013.01); *E06B 7/215* (2013.01)

(58) **Field of Classification Search**

CPC E06B 7/2312
See application file for complete search history.

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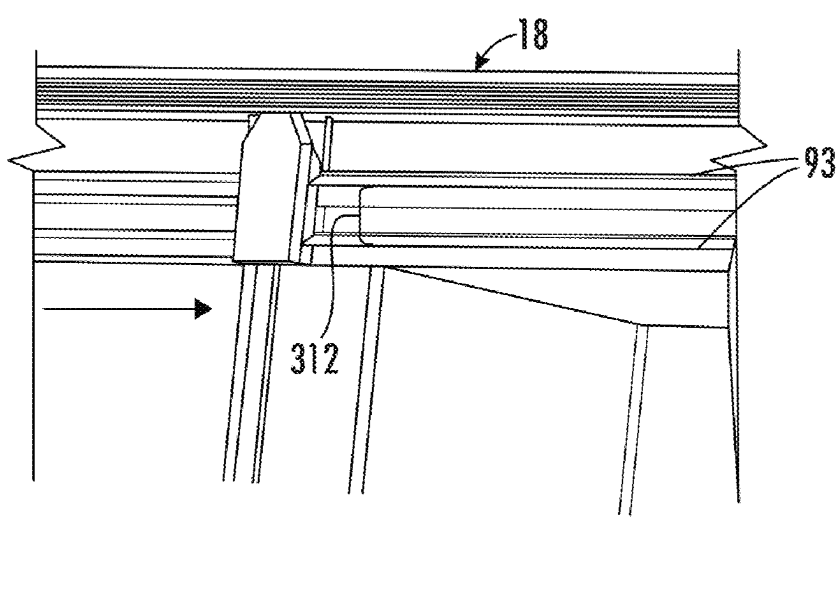
Primary Examiner — Catherine A Kelly
(74) *Attorney, Agent, or Firm* — Moore & Van Allen PLLC; Jeffrey R. Gray

(57) **ABSTRACT**

A door system includes a door having a top edge, a leading edge, a trailing edge and an inner face. A first gasket is deformed into engagement with the inner face adjacent the top edge, a second gasket is deformed into engagement with the inner face adjacent the top edge and spaced from the first gasket. An acoustic barrier is positioned to close a gap between the first gasket, the second gasket and the inner face. A third gasket is mounted to a structure adjacent the door and a fourth gasket is mounted to the trailing edge of the door. Sealing members on the first gasket engage sealing members on the second gasket when the door is in the closed position.

13 Claims, 23 Drawing Sheets



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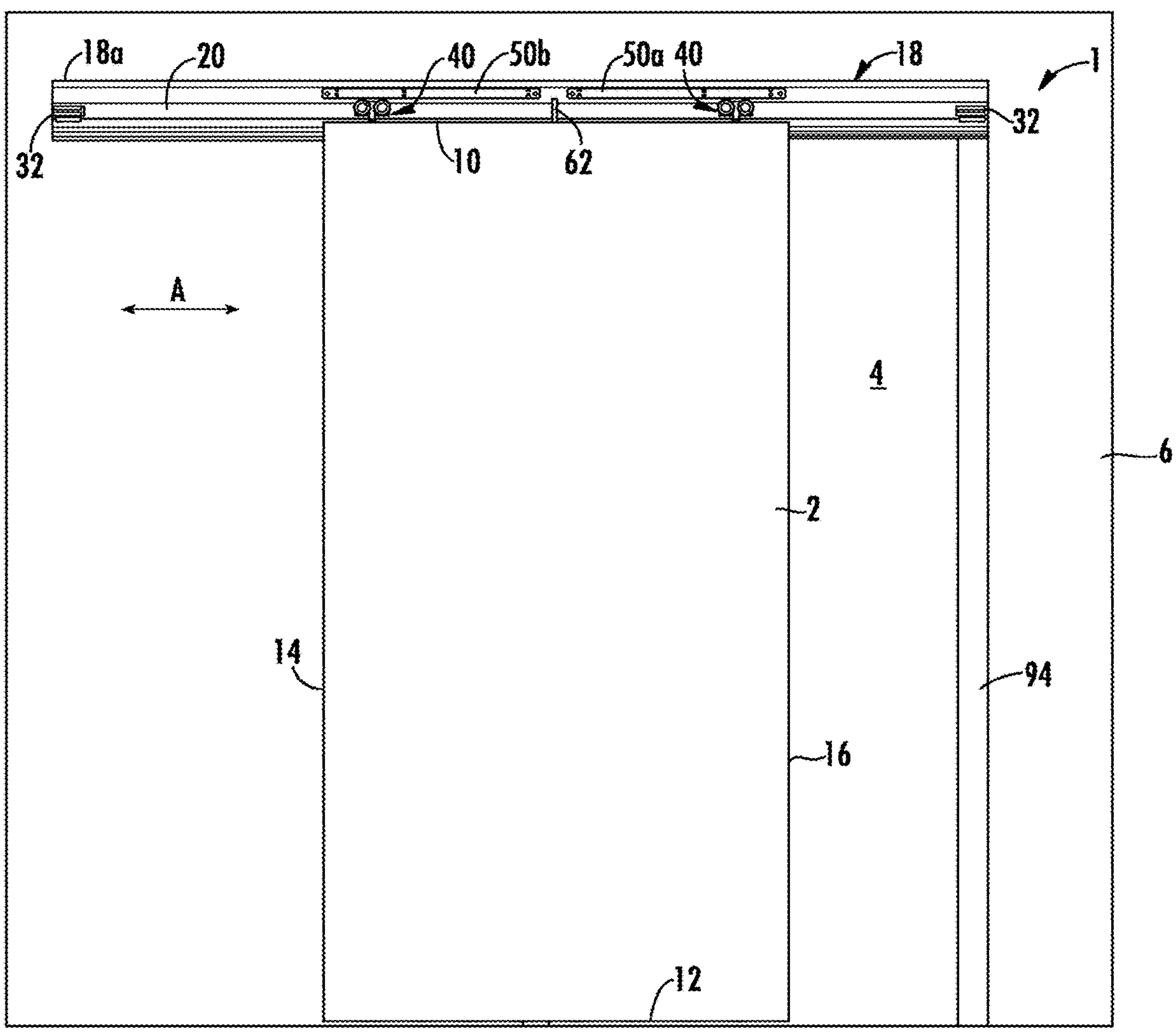


FIG. 1

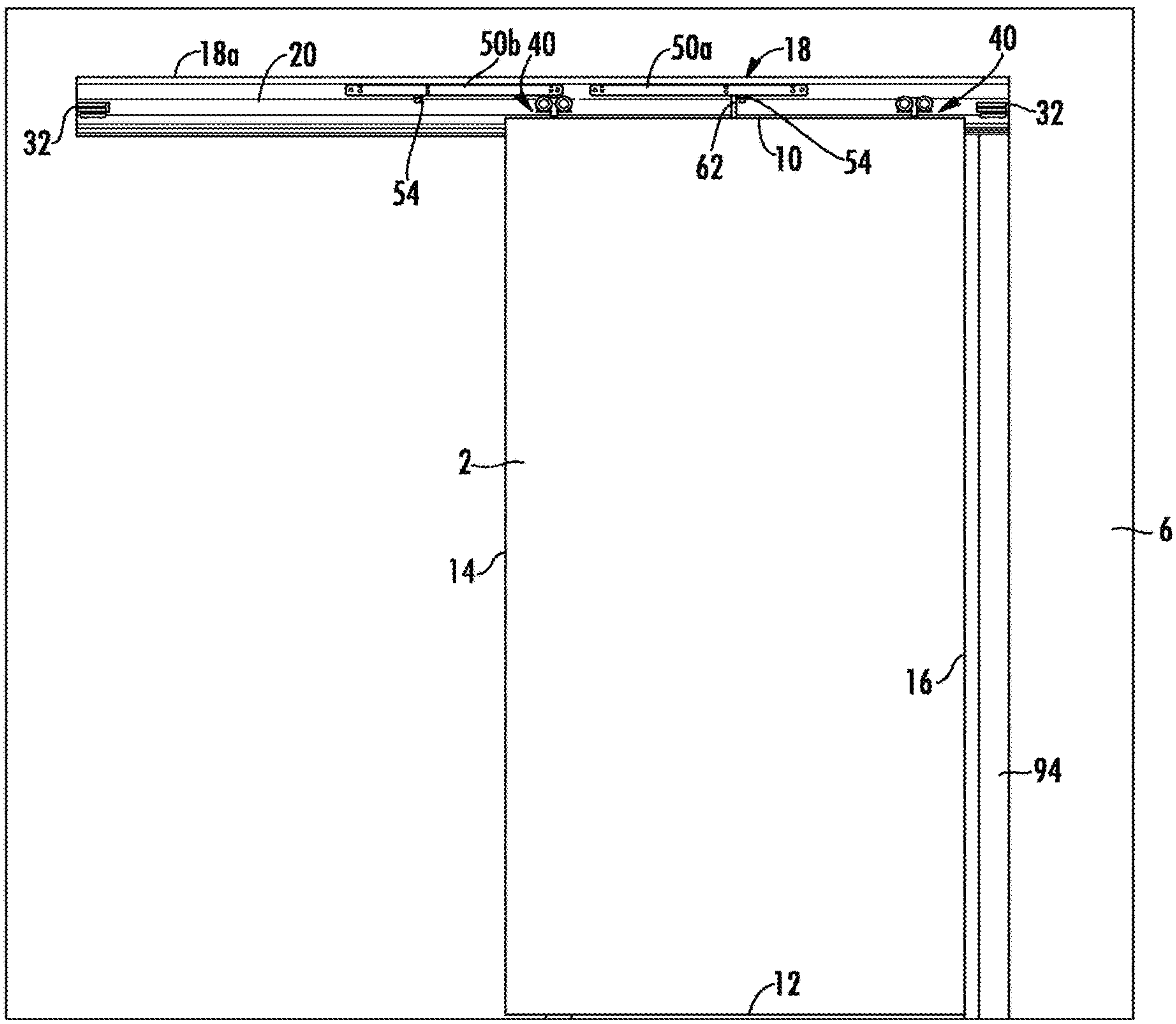
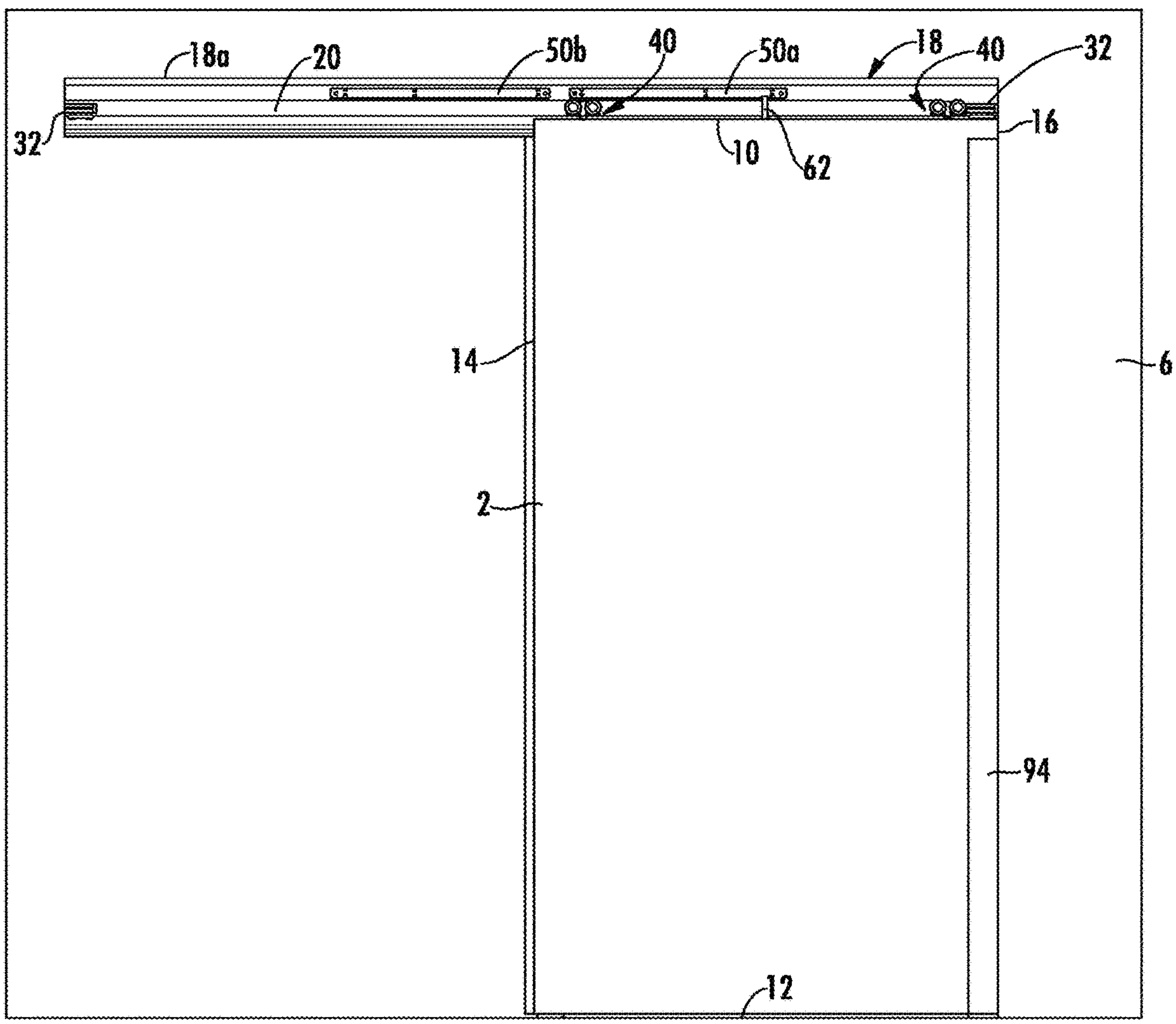


FIG. 2



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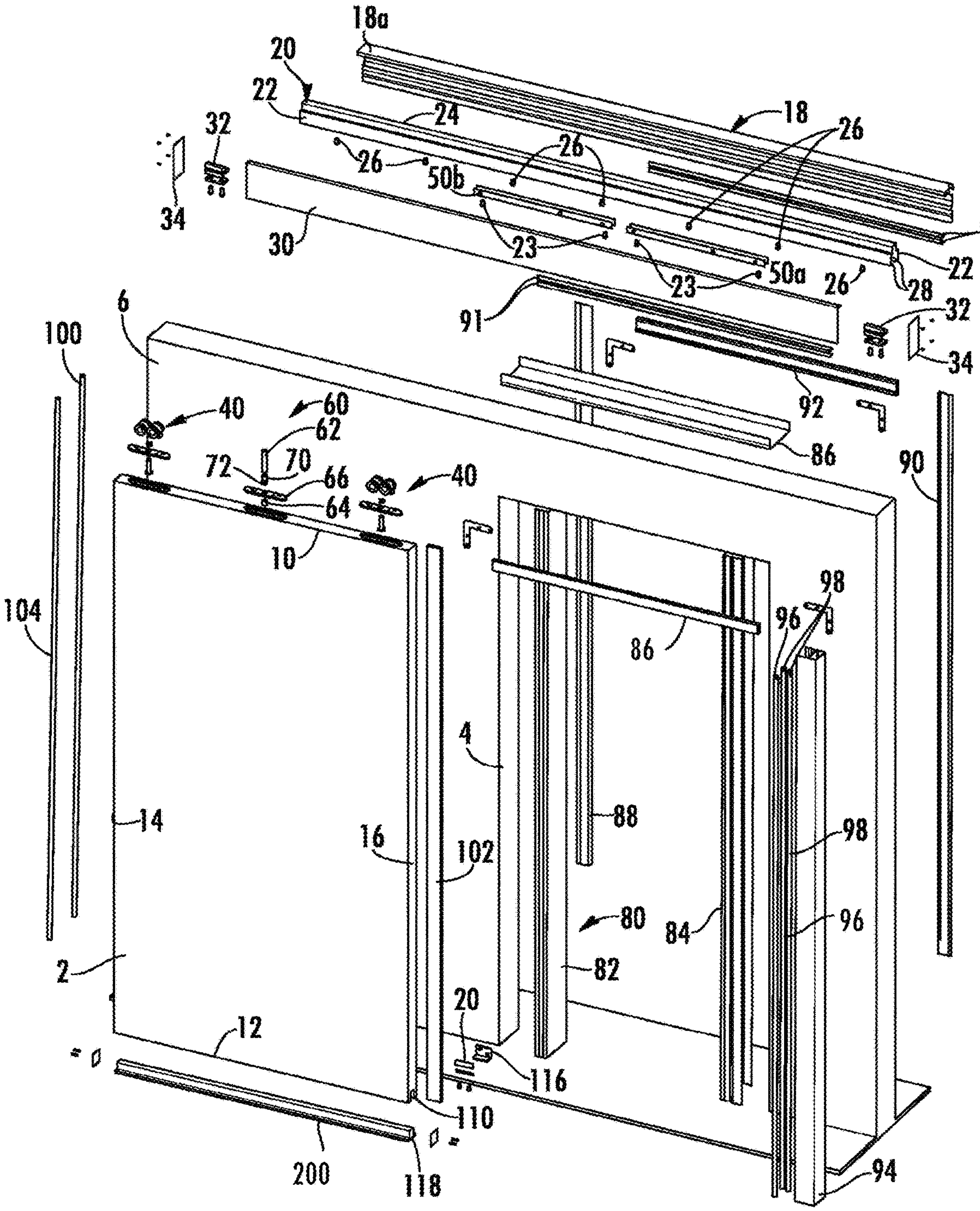
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FIG. 4

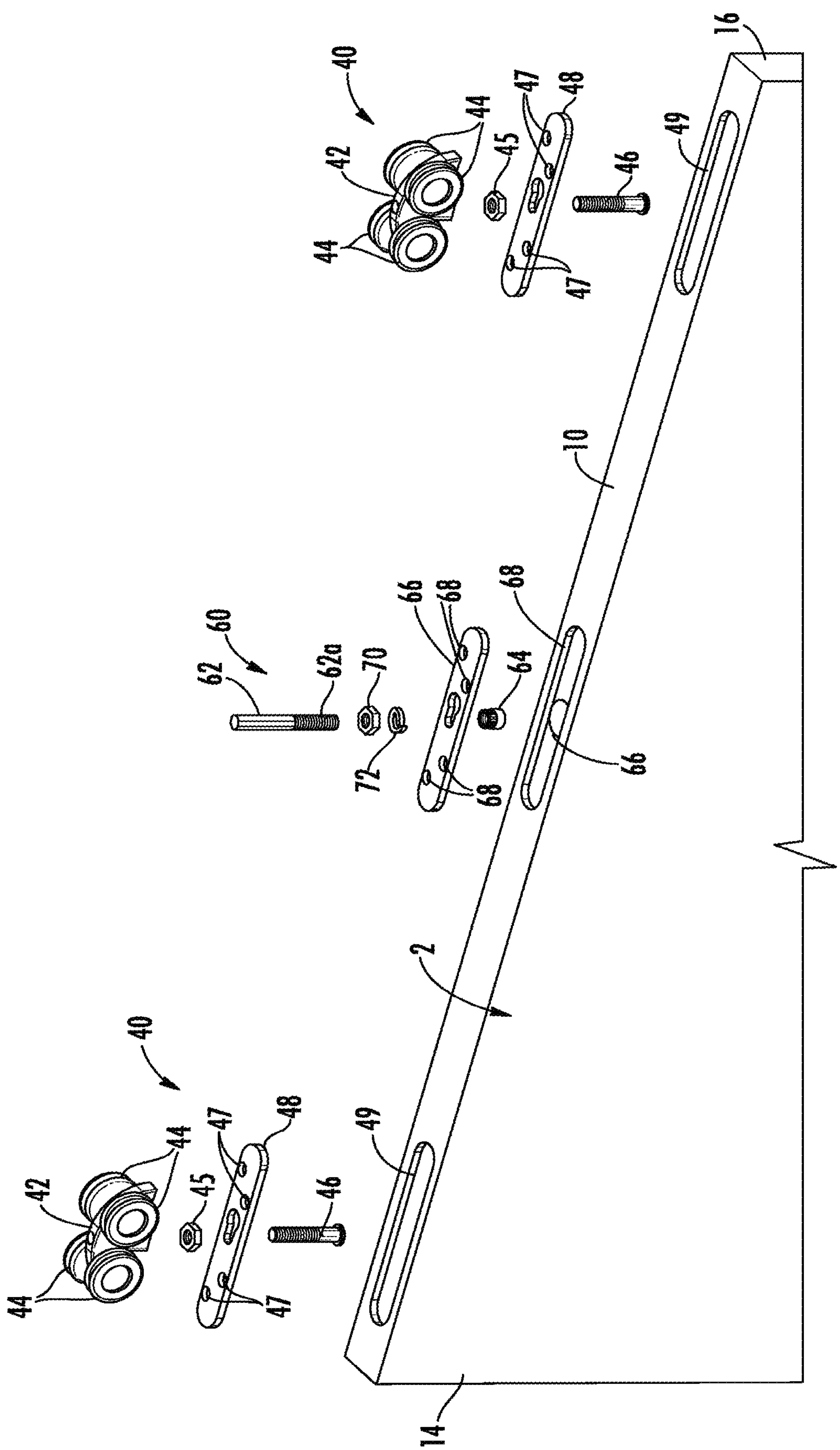
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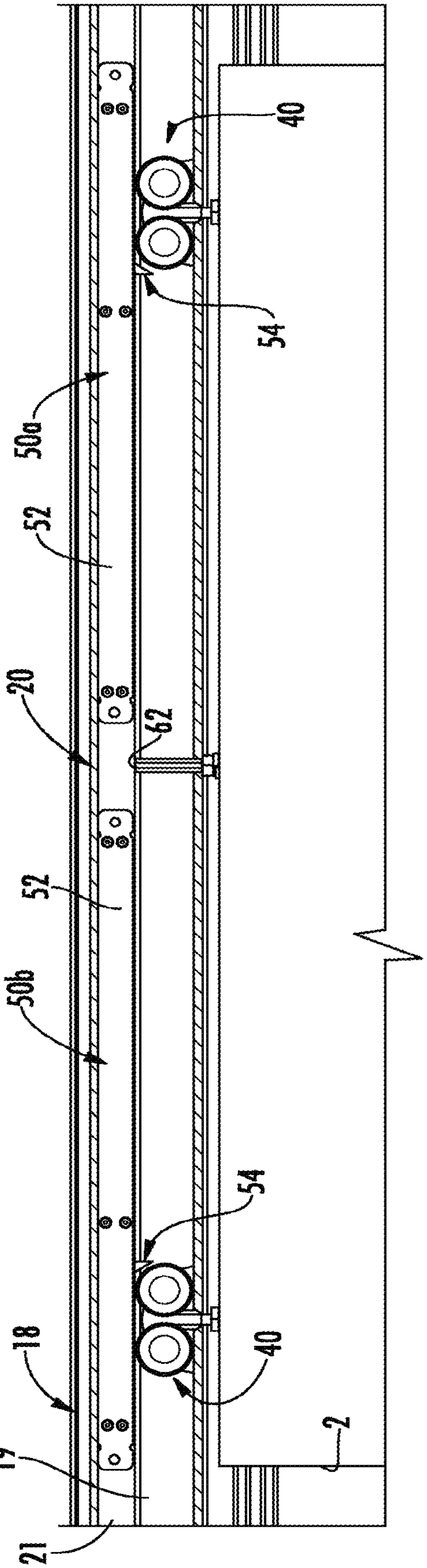
FIG. 5



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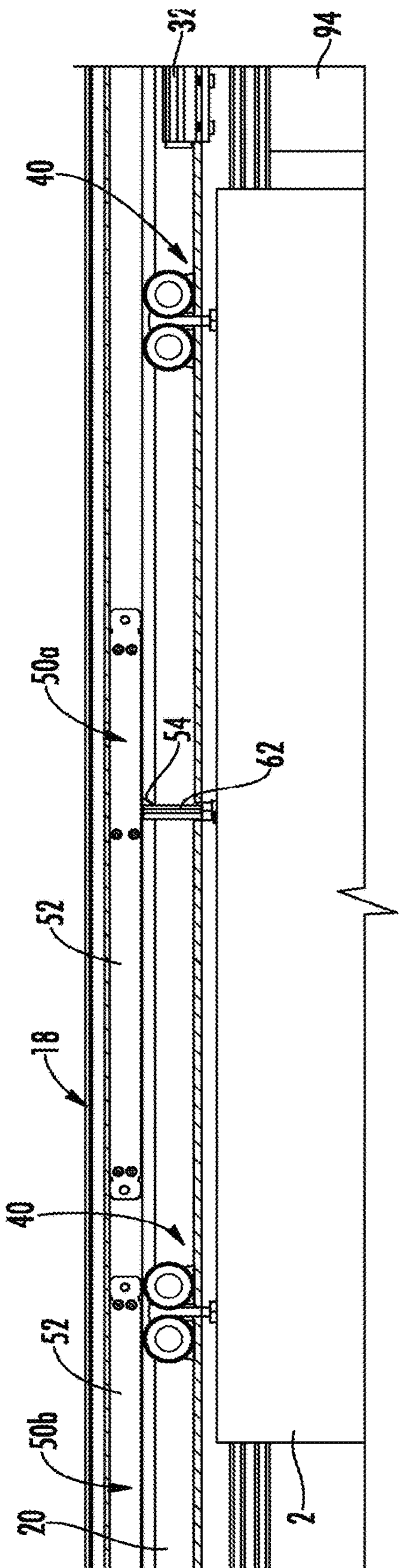


FIG. 7

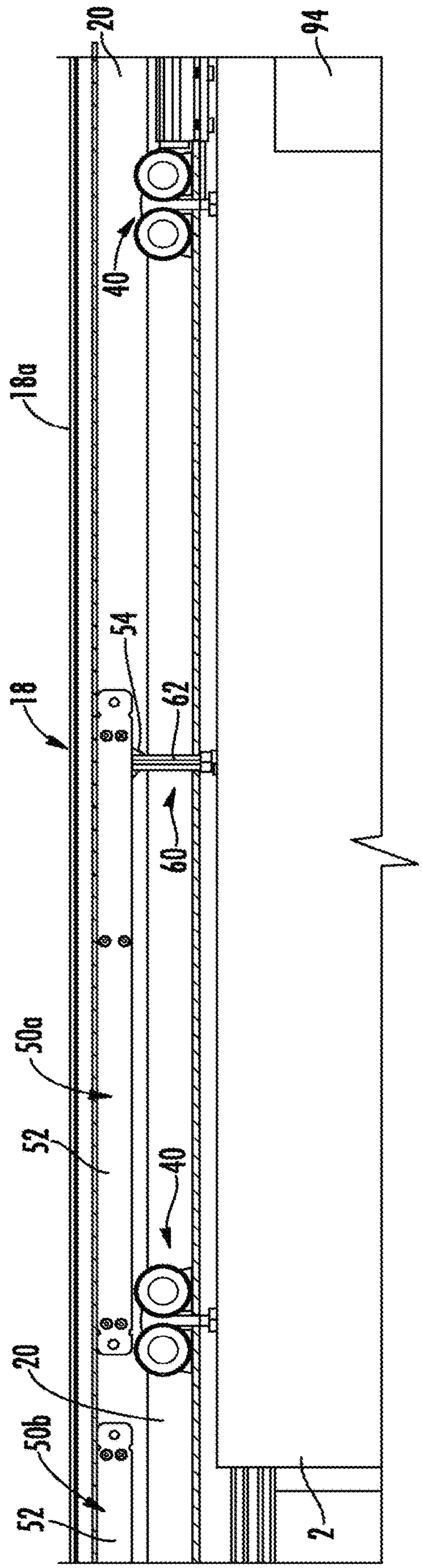


FIG. 8

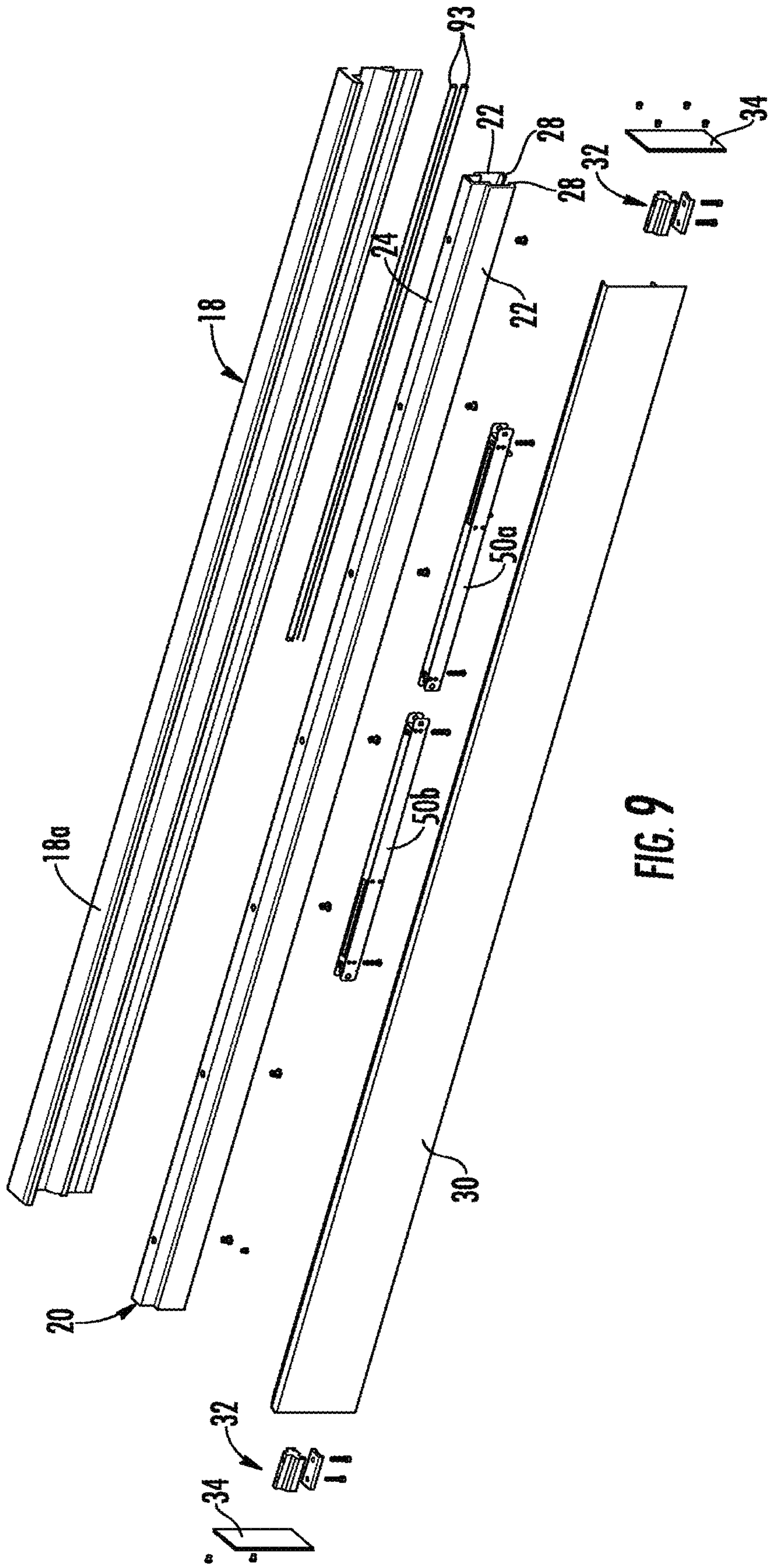
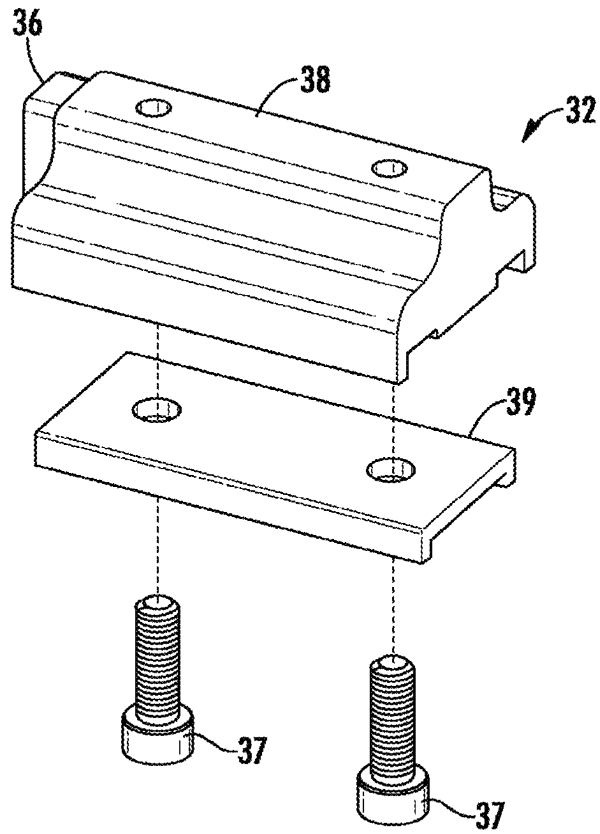


FIG. 9



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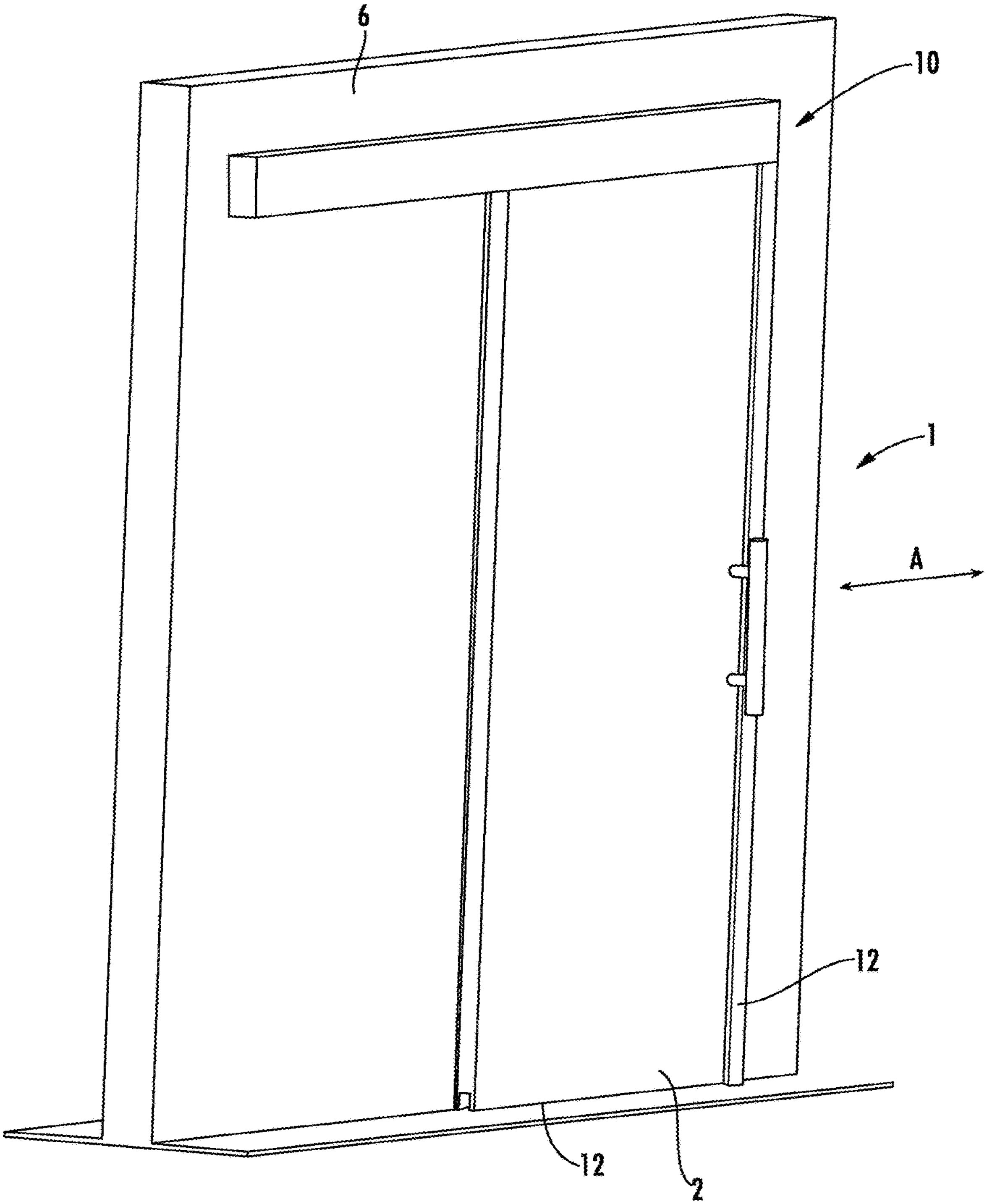

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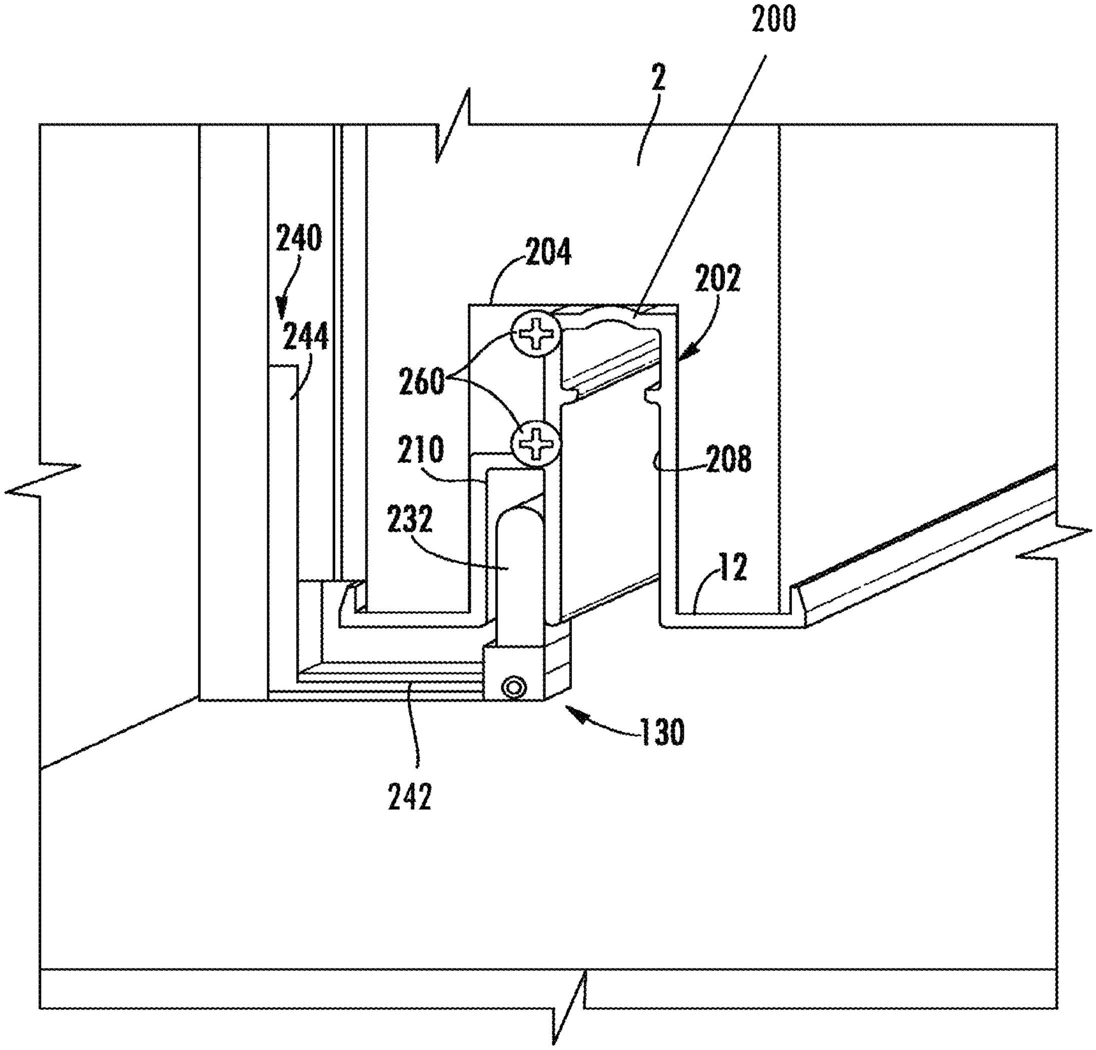

FIG. 12

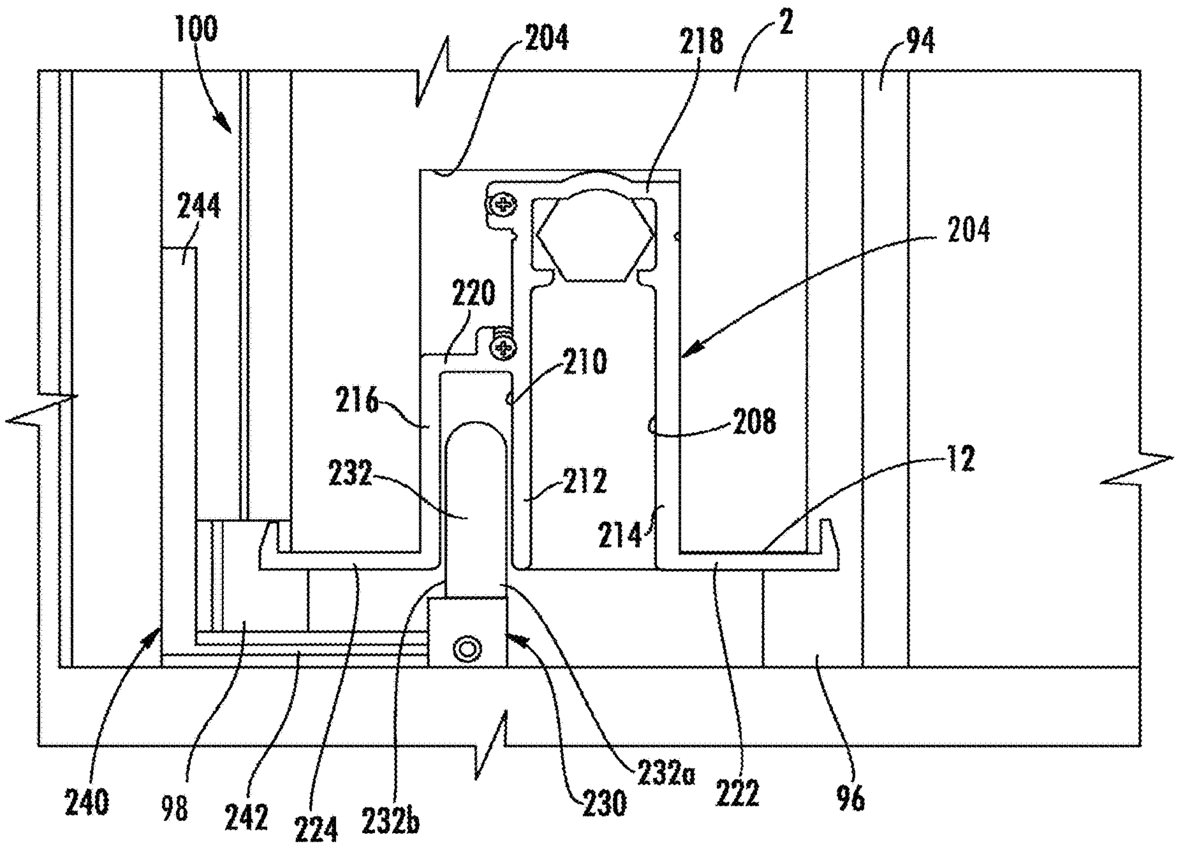
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FIG. 13

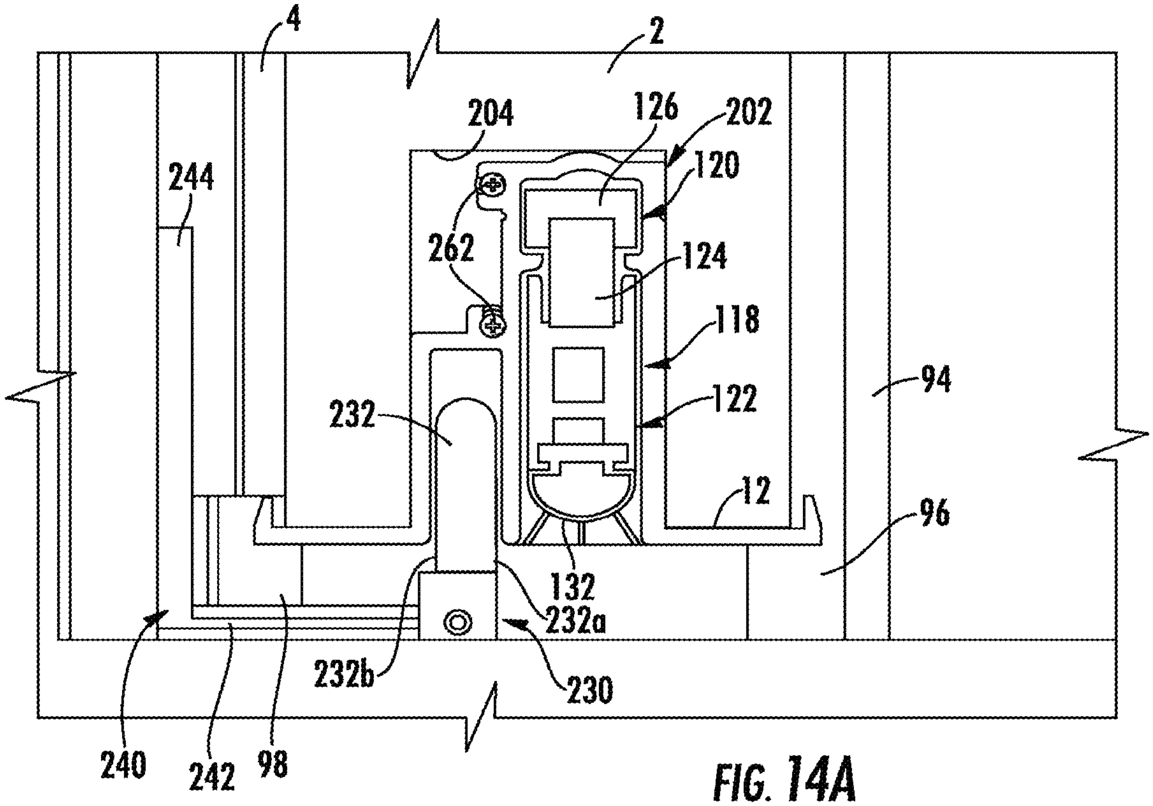
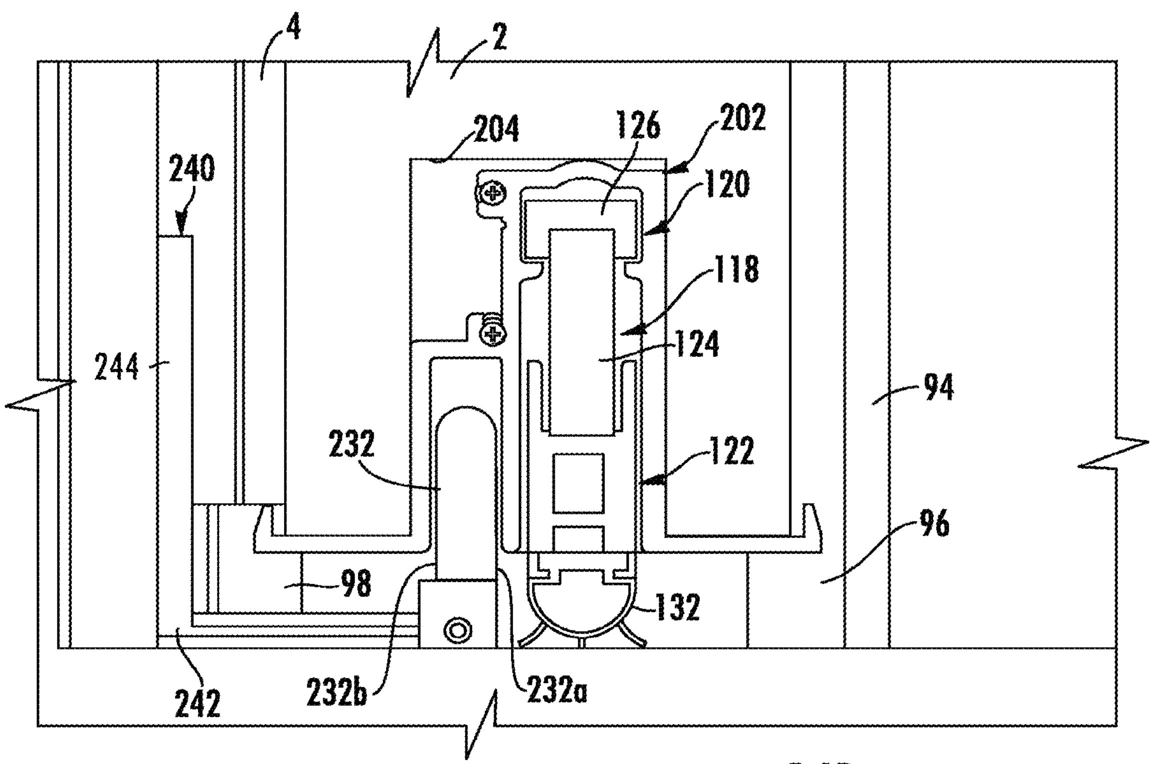


FIG. 14A



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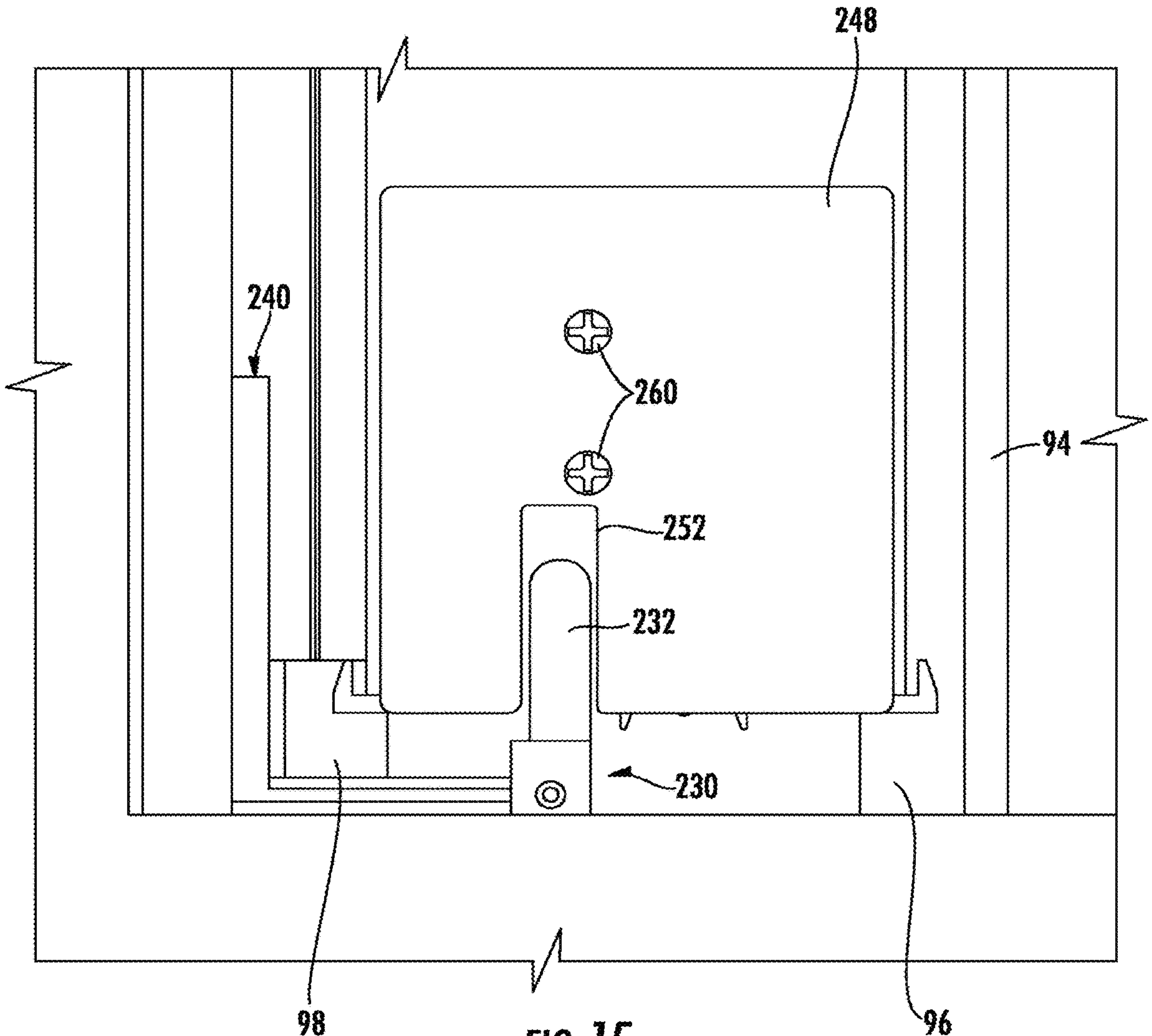

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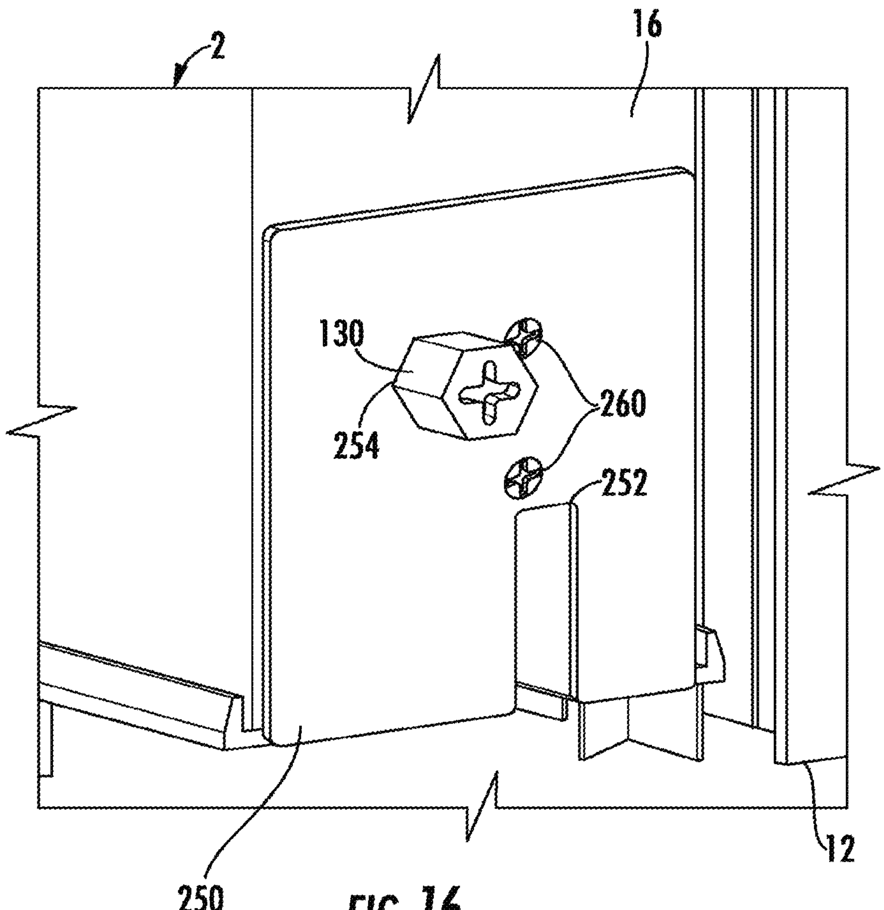

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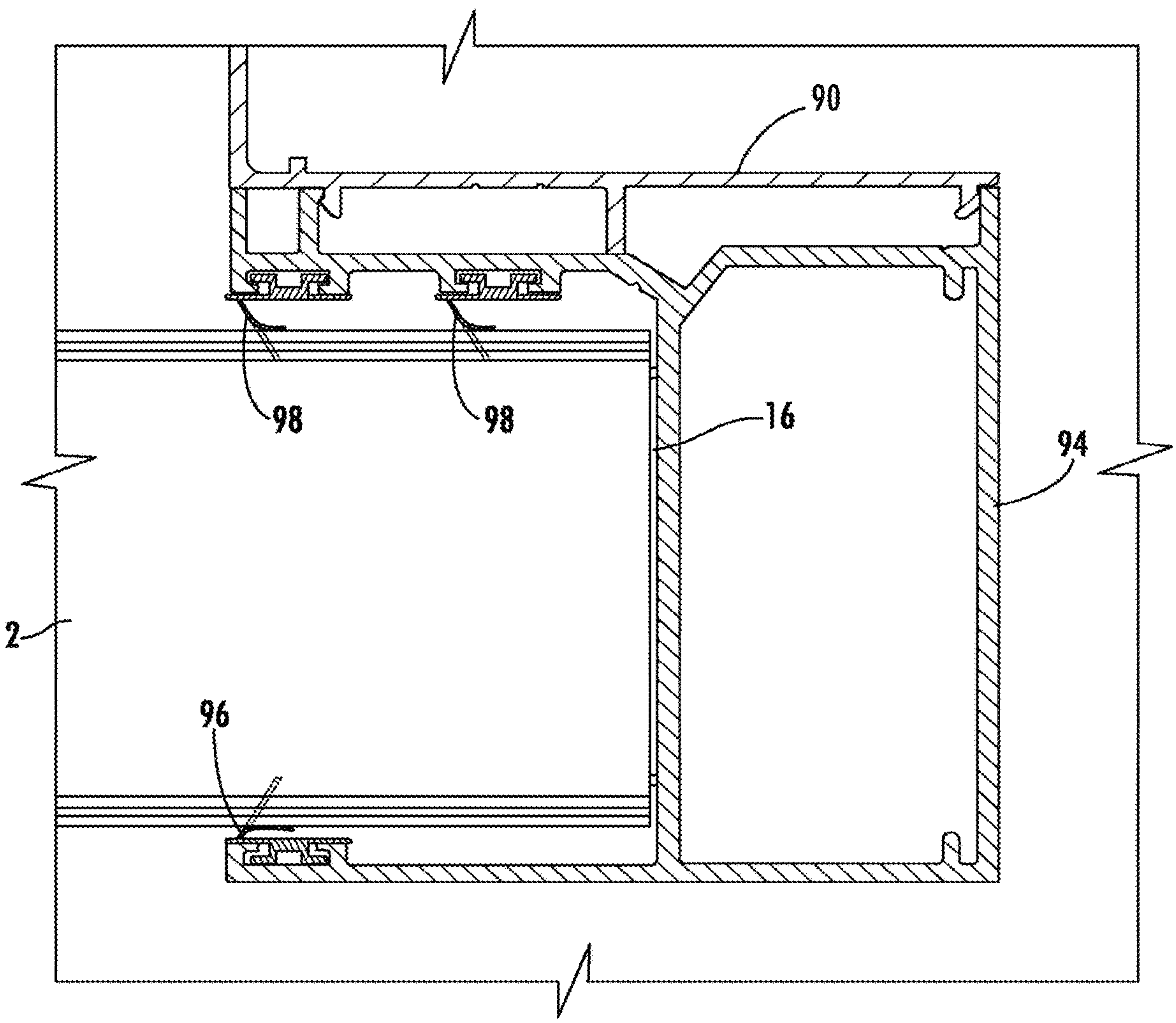

FIG. 17

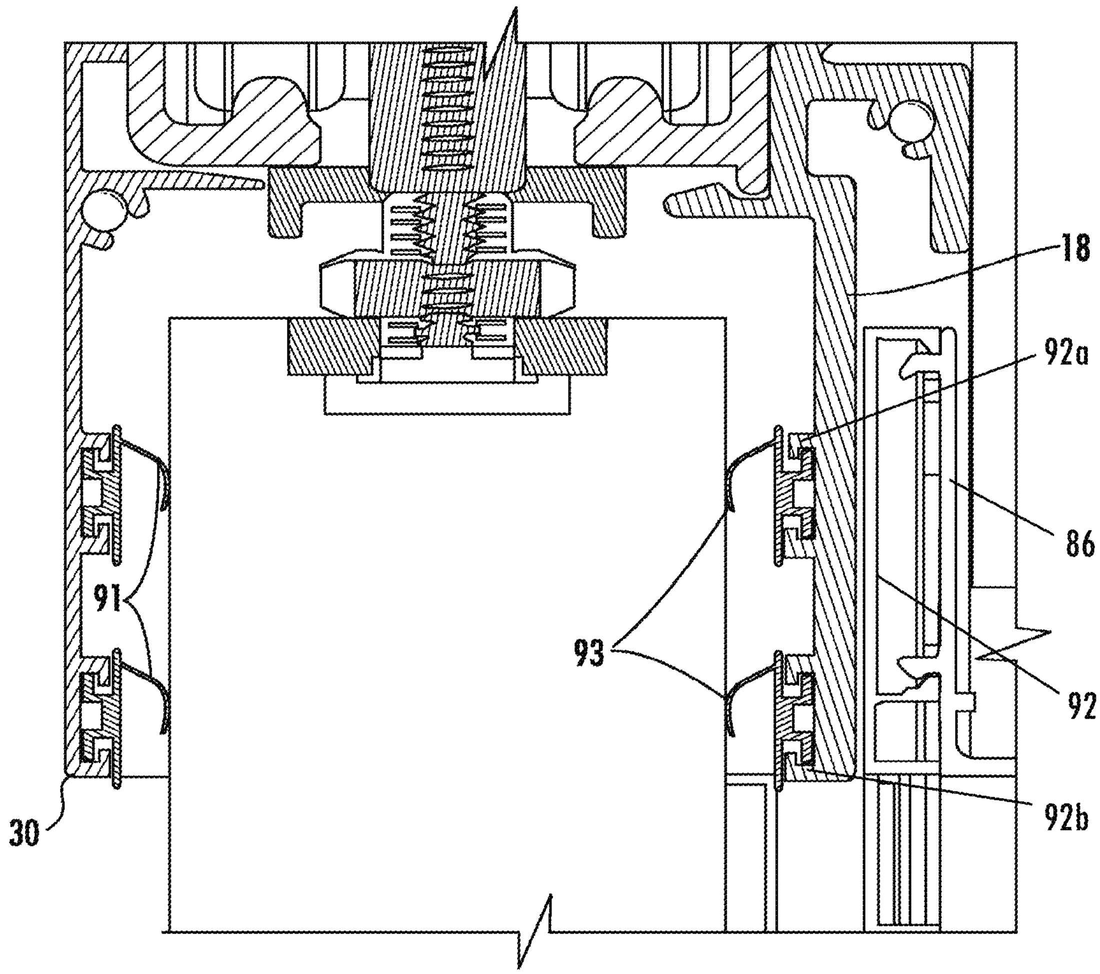

FIG. 18

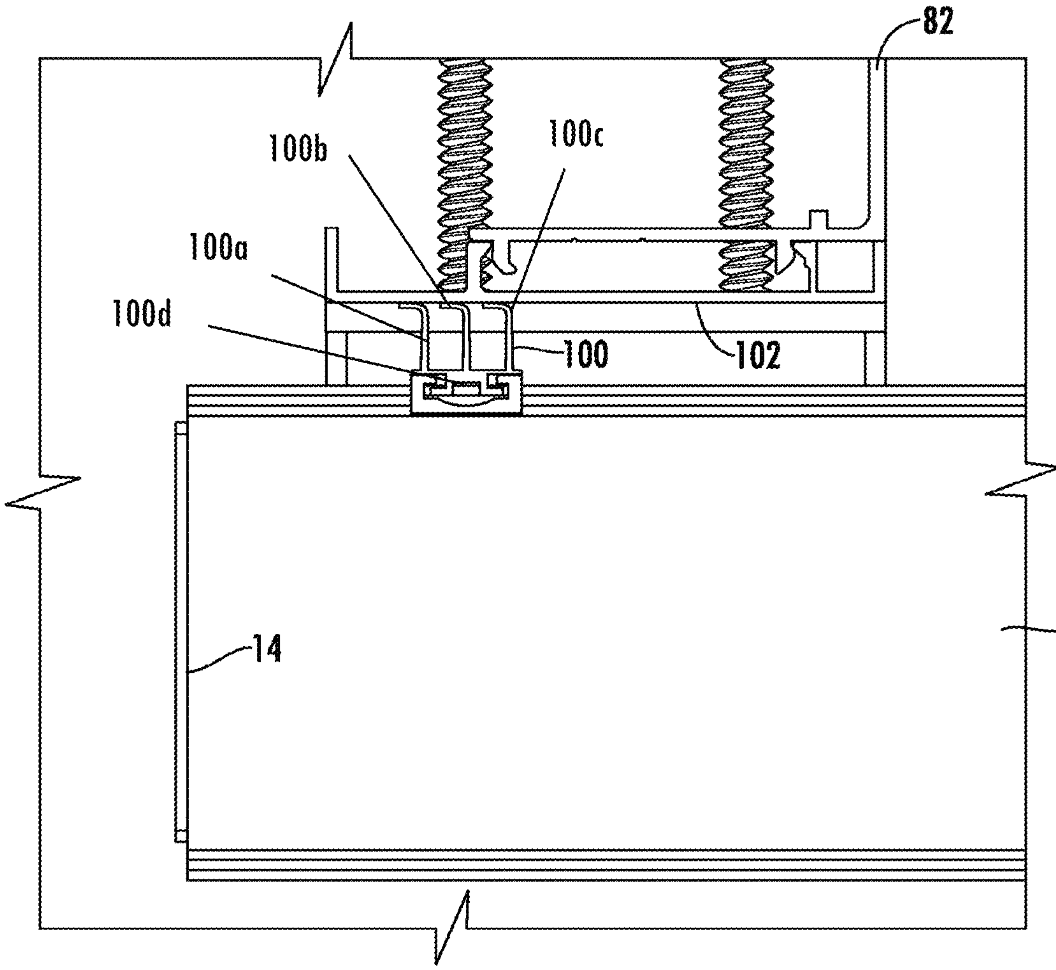

FIG. 19

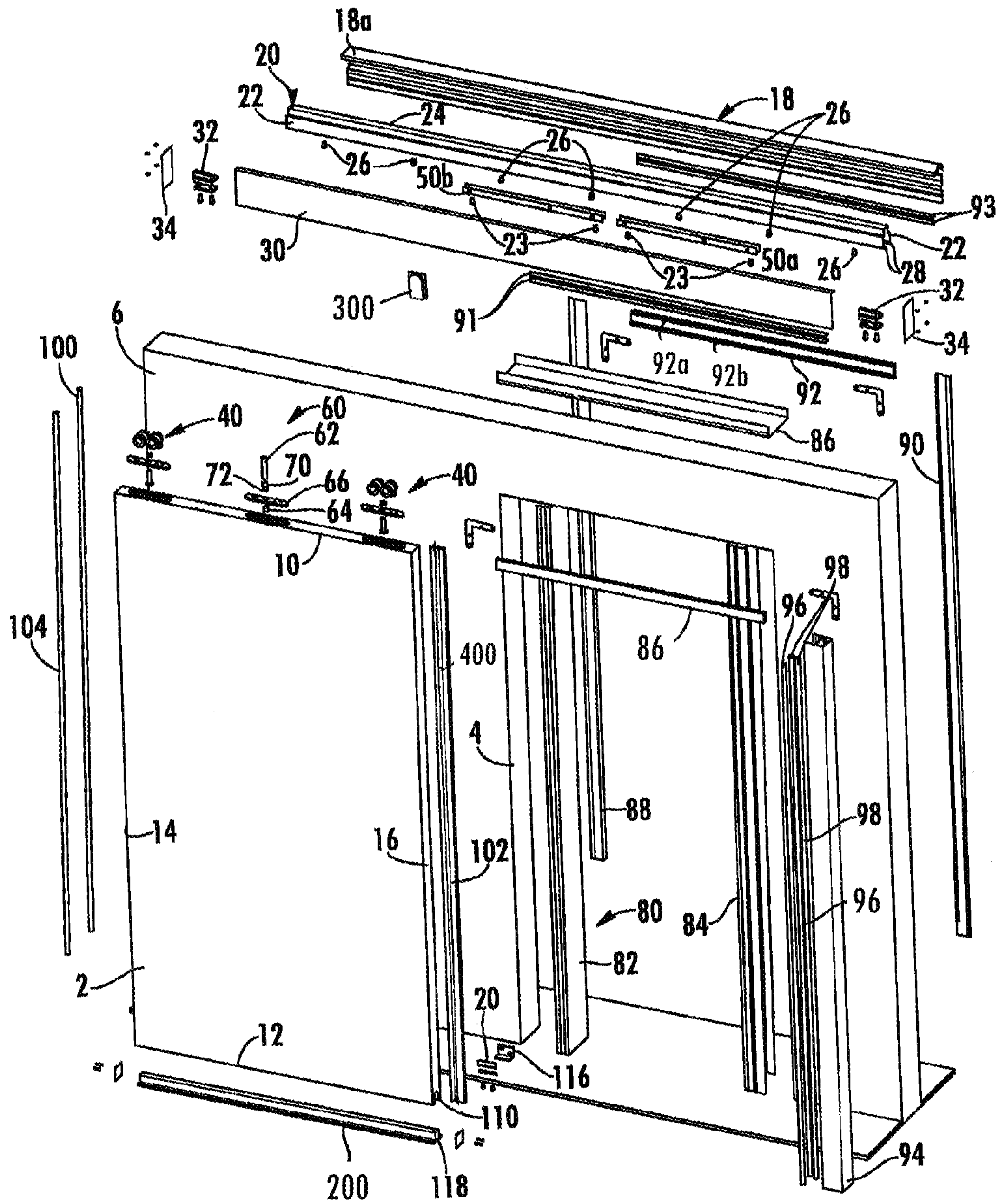
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FIG. 20

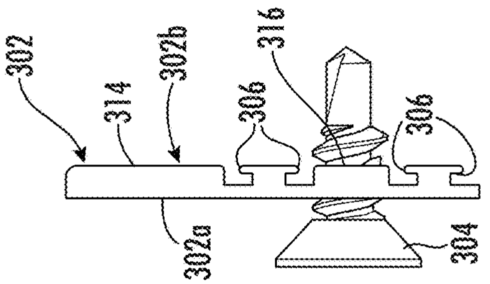

FIG. 25

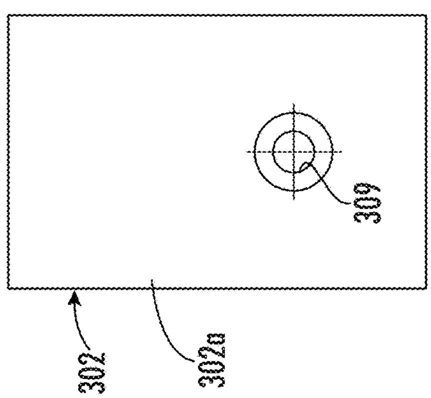

FIG. 24

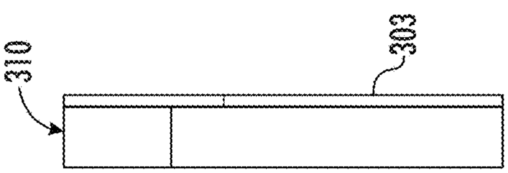

FIG. 23

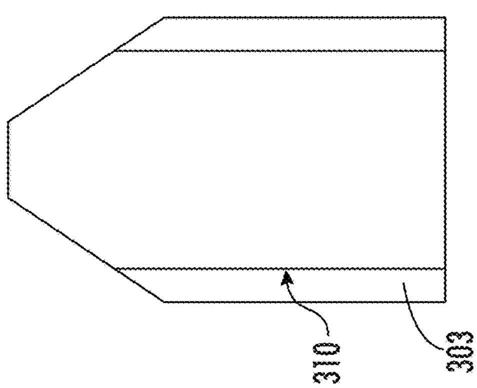

FIG. 22

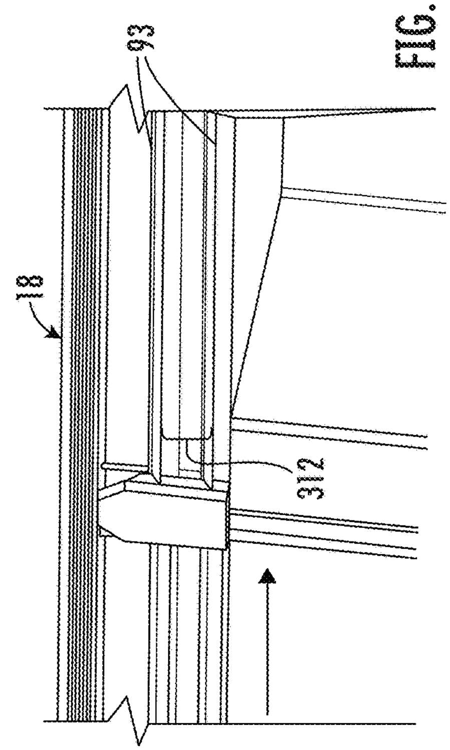

FIG. 26

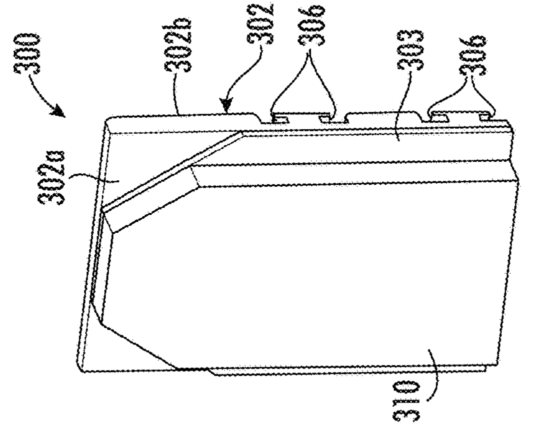

FIG. 21

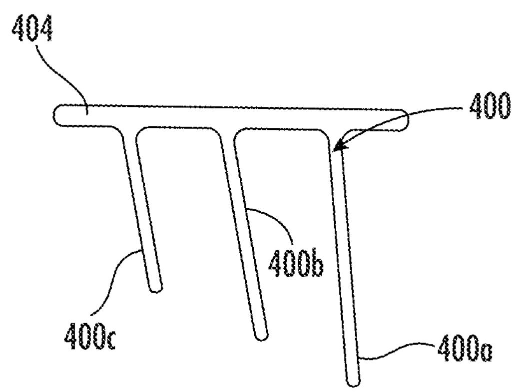

FIG. 27

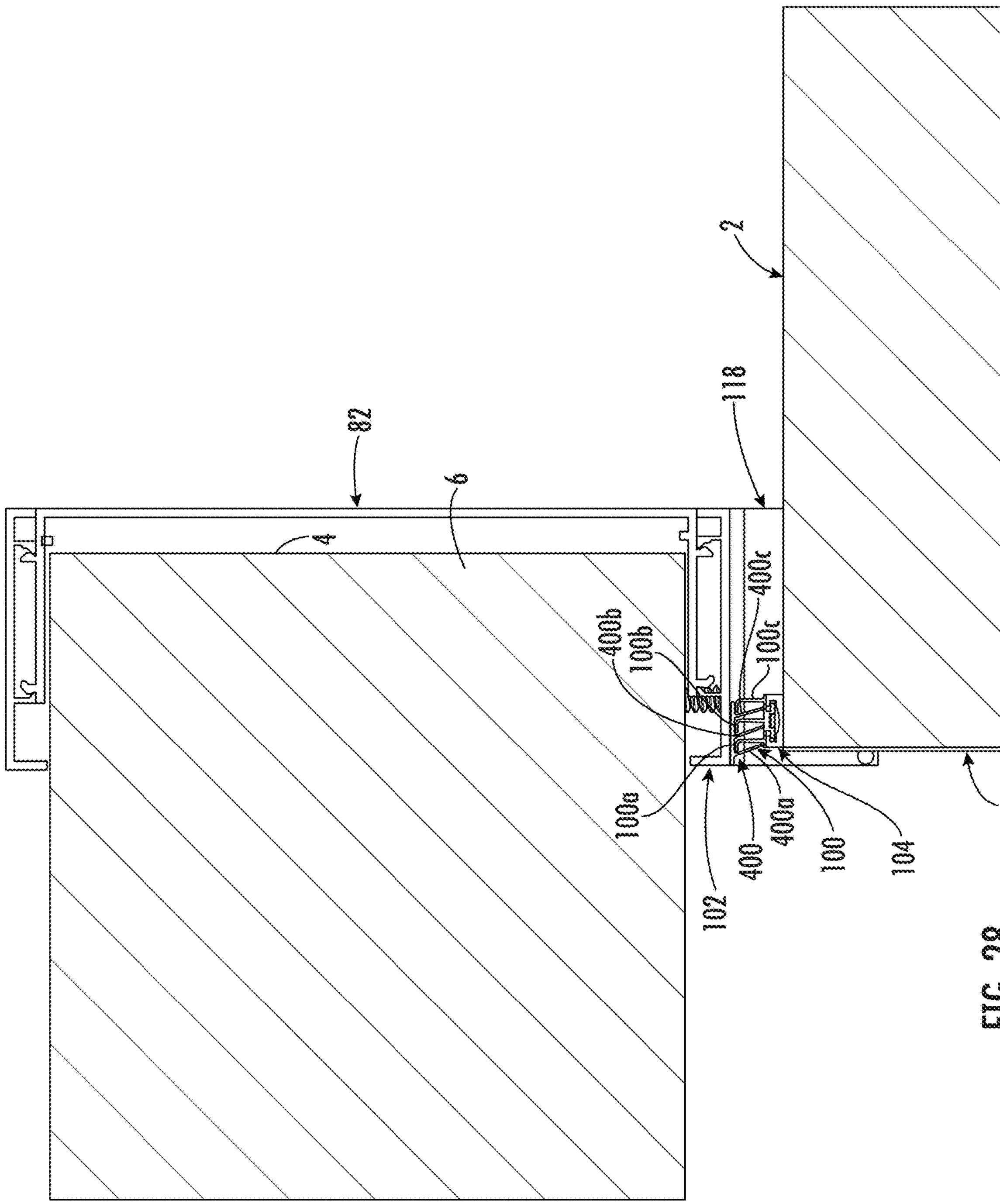

FIG. 28

INTEGRATED GUIDE SYSTEM AND DOOR SEAL FOR A SOFT CLOSE SLIDING DOOR

This application is a continuation-in-part of U.S. application Ser. No. 16/104,535, filed on Aug. 17, 2018, which claims benefit of priority under 35 U.S.C. § 119(e) to the filing date of U.S. Provisional Application No. 62/654,662, as filed on Apr. 9, 2018, both of which are incorporated by reference herein in their entirety.

FIELD

This invention relates generally to sliding doors and more particularly to soft close sliding doors and sliding doors having a sealing assembly.

BACKGROUND

A sliding door may be mounted on a track that allows the door to slide laterally relative to a door opening between an open position and a closed position. In the open position the door is spaced from the opening to allow ingress and egress through the door opening and in the closed position the door covers the opening to prevent ingress and egress through the door opening. A typical sliding door may be freely movable along the track such that the door is manually slid laterally between the open and closed positions. In certain circumstances the sliding door can approach the open and closed positions with an undesirable amount of force because the door is freely movable and the force applied to the door is variable. As a result, soft close systems have been developed that control the movement of the door as it reaches the fully open and/or fully closed positions. In some situations, it is desirable to provide a door having a seal such that when the door is in the closed position the door is sealed to prevent or substantially reduce the transmission of air, sound, debris and the like across the door opening. In some embodiments the sealing system includes a bottom seal that engages the floor when the door is in the closed position.

SUMMARY

In some embodiments, a sliding door system comprises a door comprising a top edge, a leading edge and a trailing edge defining an inner face. The door is mounted for sliding movement relative to a door opening formed in a structure between a closed positon and an open position. A first gasket and a second gasket are deformed into engagement with the inner face adjacent the top edge. The first gasket is spaced from the second gasket to create a gap between the first gasket, the second gasket and the inner face. An acoustic barrier is positioned at one end of the first gasket and the second gasket. The acoustic barrier engages the inner face and closes the gap between the first gasket, the second gasket and the inner face when the door is in the closed position.

The acoustic barrier may comprise a front side that slidably engages the inner face of the door. The front side of the acoustic barrier may be generally planar and deformable. The front side of the acoustic barrier may comprise a pad that slidably engages the inner face of the door. The acoustic barrier may comprise a plate that is dimensioned to extend between the first gasket and the second gasket. The plate may support the pad. The pad may comprise a pile pad. A back side of the acoustic barrier may comprise slots that receive a mounting fascia for the first gasket and the second gasket. The acoustic barrier may be fixed to the structure. A third gasket may be mounted to the structure such that it is positioned adjacent the trailing edge of the door when the door is in the closed position and a fourth gasket may be mounted to the trailing edge of the door. The third gasket may comprise a plurality of first deformable sealing members and the fourth gasket may comprise a plurality of second deformable sealing members. The plurality of first deformable sealing members may interdigitate with the plurality of second deformable sealing members when the door is in the closed position. The plurality of first deformable sealing members may alternate with the plurality of second deformable sealing members in a direction from the leading edge of the door toward the trailing edge of the door when the door is in the closed position. The third gasket may be mounted to the structure such that the plurality of first deformable sealing members extend at a non-perpendicular angle relative to the door. A channel member may be disposed generally vertically to receive the leading edge of the door when the door is in the closed position where the channel member comprises a fifth gasket that faces the door such that when the door is in the closed position and the leading edge of the door is received in the channel, the fifth gasket engages the door near the leading edge.

In some embodiments, a sealing assembly for a door is provided where the door comprises a top edge, a leading edge and a trailing edge defining an inner face where the door is mounted for movement relative to a door opening formed in a structure between a closed positon and an open position. A first gasket and a second gasket are deformed into engagement with the inner face adjacent the top edge where the first gasket is spaced from the second gasket to create a gap between the first gasket, the second gasket and the inner face. An acoustic barrier is positioned at one end of the first gasket and the second gasket, where the acoustic barrier engages the inner face and closes the gap between the first gasket, the second gasket and the inner face when the door is in the closed position. A third gasket is mounted to the structure such that it is positioned adjacent the trailing edge of the door when the door is in the closed position and a fourth gasket is mounted to the trailing edge of the door. The third gasket comprises a plurality of first deformable sealing members and the fourth gasket comprises a plurality of second deformable sealing members arranged such that the plurality of first deformable sealing members engage the plurality of second deformable sealing members when the door is in the closed position.

In some embodiments, a sliding door system comprises a door having a top edge, a leading edge and a trailing edge defining an inner face, the door being mounted for sliding movement relative to a door opening formed in a structure between a closed positon and an open position. A first gasket is mounted to the structure such that it is positioned adjacent the trailing edge of the door when the door is in the closed position and a second gasket is mounted to the trailing edge of the door. One of the first gasket and the second gasket comprises a plurality of first deformable sealing members and the other one of the first gasket and the second gasket comprises at least one second deformable sealing member. The plurality of first deformable sealing members interdigitate with the at least one second deformable sealing member when the door is in the closed position.

The plurality of first deformable sealing members may alternate with the at least one second deformable sealing member in a direction from the leading edge of the door toward the trailing edge of the door when the door is in the closed position. The first gasket may be mounted to the structure such that the plurality of first deformable sealing members extend at a non-perpendicular angle relative to the door. A third gasket and a fourth gasket may be deformed into engagement with the inner face adjacent the top edge. The third gasket may be spaced from the fourth gasket to create a gap between the first gasket, the second gasket and the inner face. An acoustic barrier may be positioned at one end of the first gasket and the second gasket where the acoustic barrier engages the inner face and closes the gap between the first gasket, the second gasket and the inner face when the door is in the closed position. The acoustic barrier may comprise a front side that slidably engages the inner face of the door. The front side of the acoustic barrier may comprise a soft pad. The acoustic barrier may comprise a plate that supports the pad where the plate comprises slots that receive a mounting fascia for the third gasket and the fourth gasket.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a plan view of an embodiment of a portion of a door assembly in a partially open position according to the present invention.

FIG. **2** is a plan view of the door assembly of FIG. **1** in a near closed position.

FIG. **3** is a plan view of the door assembly of FIG. **1** in a fully closed position.

FIG. **4** is an exploded perspective view of the door assembly of FIG. **1**.

FIG. **5** is a detailed exploded perspective view of the door assembly of FIG. **1**.

FIG. **6** is a detailed elevation view of the door assembly of FIG. **1** in a partially open position.

FIG. **7** is a detailed elevation view of the door assembly of FIG. **1** in a near closed position.

FIG. **8** is a detailed elevation view of the door assembly of FIG. **1** in a fully closed position.

FIG. **9** is a detailed exploded perspective view of the track and associated hardware of the door assembly of FIG. **1**.

FIG. **10** is a detailed exploded perspective view of the stop of the door assembly of FIG. **1**.

FIG. **11** is a perspective view of an embodiment of a door assembly in a closed position with an embodiment of the integrated guide assembly of the invention.

FIG. **12** is a detailed perspective view of the integrated guide assembly of FIG. **11**.

FIG. **13** is an end view of the integrated guide assembly of FIG. **11** with the bottom door seal removed.

FIG. **14**A is an end view of the integrated guide assembly of FIG. **11** with the bottom door seal in a retracted position.

FIG. **14**B is an end view of the integrated guide assembly of FIG. **11** with the bottom door seal in an extended position.

FIG. **15** is an end view of one end of the door showing the integrated guide assembly of FIG. **11** with a cover plate installed.

FIG. **16** is a perspective view of the opposite end of the door showing the integrated guide assembly of FIG. **11** with a cover plate installed.

FIG. **17** is a top view of the leading edge of the door in the closed position.

FIG. **18** is an end view of the top edge of the door.

FIG. **19** is a top view of the trailing edge of the door in the closed position.

FIG. **20** is an exploded perspective view of another embodiment of the door assembly of FIG. **1** with additional acoustic barriers.

FIG. **21** is a perspective view of the acoustic barrier of FIG. **20**.

FIG. **22** is a front view of an acoustic pad that forms part of the acoustic barrier of FIG. **21**.

FIG. **23** is a side view of the acoustic pad of FIG. **22**.

FIG. **24** is a front view of the support plate that forms part of the acoustic barrier of FIG. **21**.

FIG. **25** is a side view of the support plate of FIG. **22**.

FIG. **26** is a perspective view showing the acoustic barrier of FIGS. **20** and **21** mounted on a door.

FIG. **27** is an end view of an additional sound attenuating gasket.

FIG. **28** is a horizontal section view showing the arrangement of the sound attenuating gasket of FIG. **27** mounted on the trailing edge of the door in the closed position.

DETAILED DESCRIPTION

Embodiments of the present invention now will be described more fully hereinafter with reference to the accompanying drawings. Referring to the figures, an embodiment of a sliding door assembly is shown. While one embodiment of a door assembly is shown and described, the inventions described herein may be used in door assemblies other than the embodiment specifically shown and described herein.

The door assembly is shown generally at **1** and comprises a door **2** mounted for movement relative to a door opening **4** formed in a wall or other structure **6**. The door opening **4** may provide ingress to or egress from a room, building, or other structure. The opening **4** may be selectively opened or closed by moving the door **2** laterally relative to the opening **4** between a fully open position and a fully closed position as represented by arrow A in FIG. **1**. It will be appreciated that the door may assume partially open/closed positions between the fully open position and the fully closed position.

The door **2** may comprise a generally planar member defining a top edge **10**, bottom edge **12**, a first or trailing edge **14** and a second or leading edge **16**. The door may be made of any suitable material and may be provided in a variety of suitable finishes and styles to meet both aesthetic needs and utilitarian requirements such as sound, smoke, fire proofing or the like. To support the door **2** for movement relative to the wall **6**, an elongated mounting bracket **18** is mounted to the wall **6** over the opening **4**. The mounting bracket **18** is mounted substantially horizontally and may extend for the full length of travel of the door **2**. The mounting bracket **18** may be secured to the wall **6** by fasteners such as screws or the like. A track **20** is mounted to the mounting bracket **18** such that it extends for the full length of travel of the door **2**. In one embodiment, the mounting bracket **18** has an overhang portion **18*a***. The track **20** is mounted substantially horizontally and extends over the opening **4** such that it supports the door **2** between the fully open position and the fully closed position. The mounting bracket **18** and track **20** may be made of a rigid, strong material such as, but not limited to, metal such as aluminum, steel, or the like, a plastic or polymer material or the like, or other suitable material, provided that the mounting bracket **18** and track **20** can support the door **2**. The track **20** has a generally inverted U-shape cross section that includes downwardly extending support members **22** connected at their upper ends by a cross member **24**. The cross member **24** may be secured to the overhang portion **18*a*** of the support track **18** by fasteners **26** such as screws. The lower ends of the support members **22** terminate in inwardly extending support surfaces **28** that define a gap therebetween. Trollies **40** are supported on the support surfaces **28** and extend into the gap, as will herein after be described.

The support members **22** are formed to define a generally rectangular lower interior space **19** that communicates with the gap formed between the inwardly extending support surfaces **28** and a generally rectangular upper interior space **21** that communicates with the lower interior space, see FIGS. **1** and **6**. The upper interior space **21** is dimensioned to retain the soft close devices **50***a*, **50***b* and the lower interior space is dimensioned to receive the trollies **40** and the stops **32** as will be hereinafter explained.

A front plate **30** may be secured to the mounting bracket and/or to the track to cover the support assembly and hide the internal components. The front plate **30** may be made of any suitable material and may be snap-fit to the mounting bracket **18** or it may be secured by separate fasteners. The front plate **30** may be provided in a variety of suitable finishes and styles to meet both aesthetic and utilitarian requirements.

A stop **32** may be secured to and supported at each end of the track **20** to close the track after the trollies **40** and soft close devices **50** are inserted into the track. Referring to FIG. **10** each stop **32** may comprise an elastomeric bumper or cushion **36** supported in a block **38** such that the bumper **36** extends from the block in a position where the trolley **40** may strike the bumper **36**. Threaded members **37** threadably engage the block **38** and reverse threadably engage an opposing mounting bracket **39** such that rotation of the threaded members **37** forces the block **38** and plate **39** together. Block **38** is positioned in the lower interior space and plate **39** is positioned outside of and below the track **20** such that when the stops **32** are disposed in the track **20** and the block **38** and plate **39** are moved toward one another the support surfaces **28** of track **20** are clamped between the block **38** and plate **39**. The stops **32** may be secured in the track by any suitable mechanism. An end plate **34** may be attached to each end of the mounting bracket **18** and front plate **30** such as by screws to complete the assembly. The end plates **34** may be provided in a variety of suitable finishes and styles to meet both aesthetic needs and utilitarian requirements such as sound, smoke, fire proofing or the like

A frame **80** may be provided that defines or surrounds the opening **4**. The frame **80** may be formed by individual frame members or caps **82**, **84**, **86** that fit over the opening **4** formed in wall **6** where the caps **82**, **84**, **86** may be secured to the wall by suitable fasteners. A greater or fewer number of frame members may be used. In other embodiments the frame **80** may be formed as part of the wall **6**. The frame **80** may also comprise frame facias **88**, **90**, **92**, **102** that connect to the caps using a snap-fit connection separate fasteners or the like.

In some embodiments, a seal between the door **2** and the wall **6** and door frame **4** is provided. The seal may create a barrier against sound, air, smoke, light, sightlines, debris or the like. Gaskets **91**, **93**, **96**, **98**, and **100** are provided to create the seal. The gaskets **91**, **93**, **96**, **98** and **100** may be formed of any suitable compressible, resilient material such as rubber, polymer or the like. The gaskets **91**, **93**, **96**, **98** and **100** create a seal around the periphery of the opening **4**. In some embodiments each gasket **91**, **93**, **96**, **98** and **100** may comprise at least one elongated sealing member that extends to the door to form a seal against the door. Referring to FIG. **18**, gaskets **91** are supported in front plate **30** such that they extend along and engage the outer face of the door adjacent the top edge **10** of the door when the door is in the closed position. The gaskets **91** engage the front face of the door and are deformed into engagement with the front face to create the seal. The gaskets **91** may be supported in elongated slots or holders formed on the front plate **30** or by other securement mechanisms. Gaskets **93** are supported on mounting bracket **18** such that they extend along and engage the inner face of the door adjacent the top edge **10** of the door when the door is in the closed position. The gaskets **93** engage the inner face of the door and are deformed into engagement with the inner face to create the seal. The gaskets **93** may be supported in elongated slots or holders formed on the mounting bracket **18** or by other securement mechanisms.

Referring to FIGS. **20** through **26**, in some embodiments, an acoustic barrier **300** may be provided to increase the sound proofing of the door. The embodiment of FIG. **20** uses the same reference numbers to describe the same elements previously described with reference to FIGS. **1-19**. The acoustic barrier **300** prevents sound from penetrating the door through a gap **312** (FIG. **26**) formed between the vertically spaced gaskets **93** and the inner face of the door **2** when the door is in a closed position.

The acoustic barrier **300** comprises a relatively flat plate **302** that is dimensioned to fit between the door **2** and the structure **6** and to extend across the gaskets **93**. A front side **302***a* of plate **302** is planar and faces the door **2** of the plate. A pad **310** is attached to the front side **302***a* of the plate **302** such as by an adhesive layer **303**. The pad **310** is deformable and may be resiliently deformable such that it may be pressed into engagement with the inner face of the door **2**. The pad **310** may be formed of a relatively soft material such as a pile material.

A back side **302***b* of the plate **302** is also generally planar and faces the support structure **6**. Two pairs of grooves or slots **306** are formed in the back side **302***b* of the plate **302** and extend through the width of the plate from the leading edge to the trailing edge of plate **302**. The grooves or slots **306** extend parallel to and are aligned with the gaskets **93**. In a typical installation, the direction of travel of the door is generally horizontal such that the gaskets **93** and grooves or slots **306** extend substantially horizontally. The grooves or slots **306** are shaped and dimensioned to closely but slidably receive flanges **92***a*, **92***b* on the bracket **18** for the gaskets **93** such that the plate **302** may be slid onto the end of the bracket **18**. In the illustrated embodiment, the grooves or slots **306** each define a pair of parallel substantially L-shaped recesses that are shaped and dimensioned to receive matingly shaped flanges **92***a*, **92***b* on the bracket **18**. The back side **302***b* of plate **302**, except for grooves or slots **306**, is planar and is configured such that it engages the structure **6** between and immediately adjacent to the gaskets **93**. In the illustrated embodiment, portion **314** of back side **302***b* engages bracket **18** in the area immediately above the gaskets **93**. Portion **316** of back side **302***b* engages the bracket **18** in the area between the gaskets **93**. Portion **318** of back side **302***b* engages bracket **18** in the area immediately above the gaskets **93**. The plate **302** is fixed to the structure **6** such as by a screw **304** that engages aperture **309** on plate **302** or other fixing mechanism such that the acoustic barrier **300** is fixed in position as shown in FIG. **26**. The acoustic barrier **300** is positioned just beyond the end of the gaskets **93** at the trailing edge **14** of the door **2**. The plate **302** is configured such that the back side **302***b* of the plate, except for the area of grooves or slots **306**, is planar and abuts the structure **6**. The pad **310** contacts the back side of the door **2** such that door **2** slides over pad **310** as the door is opened and closed. The acoustic barrier **300** closes gaps that would otherwise be found at the end of gaskets **93** to further attenuate the passage of sound through the door opening.

Referring to FIG. **19** gasket **100** is mounted to the back face of the door and extends for substantially the entire height of the door. The gasket **100** engages the facing surface of the wall **6** or fascia **102**, as shown, and is deformed into engagement with the wall **6** or fascia **102** to create the seal along the trailing edge **14** of the door **2**. The gasket **100** may be supported in elongated slots or holder **104** formed on or mounted to the door or by other securement mechanisms.

Referring to FIGS. **20**, **27** and **28**, to further enhance the sound attenuation of the door, a mating second gasket **400** is mounted on the door frame **4**. The mating second gasket **400** may be mounted to fascia **102**, as shown in FIG. **20**, and extends for substantially the entire height of the door **2**. The mating second gasket **400** is located along the trailing edge **14** of the door **2** when the door is in the closed position. The gasket **400** is mounted in a fixed position on structure **6** such that the door **2** slides over the gasket **400** as the door **2** is opened and closed. In one embodiment, the gasket **400** is shown in end view in FIG. **27** and comprises three sealing members **400***a*, **400***b* and **400***c* that extend from a base member **404**. The gasket **400** may be formed of any suitable compressible, resilient material such as rubber, polymer or the like. The three sealing members **400***a*, **400***b* and **400***c* extend at an angle relative to the base **404** such that when the base **404** is secured to the structure **6** the sealing members **400***a*, **400***b* and **400***c* angle or tilt in a direction toward the leading edge **16** of the door **2** and away from the trailing edge **14** of the door **2**. The sealing members **400***a*, **400***b* and **400***c* also decrease in length from base **404** from the leading edge to the trailing edge.

The sealing members **400***a*, **400***b* and **400***c* are configured and spaced from one another such that when the door is in the closed position, the sealing members **400***a*, **400***b* and **400***c* interdigitate with the sealing members **100***a*, **100***b* and **100***c* of gasket **100**. The sealing members **400***a*, **400***b* and **400***c* alternate with the sealing members **100***a*, **100***b* and **100***c* in a direction from the leading edge of the door toward the trailing edge of the door when the door is in the closed position. Specifically, sealing member **400***c* is positioned outside of sealing member **100***a*, sealing member **400***b* is positioned between sealing member **100***a* and **100***b* and sealing member **400***c* is positioned between sealing member **100***b* and **100***c*. As shown in FIG. **28**, the sealing members **100***a*, **100***b*, **100***c* and the sealing members **400***a*, **400***b*, **400***c* are not shown in a deformed state as would be the case in actual use of the door where the sealing members **400***a*, **400***b*, **400***c* are compressed and deformed against the door **2** and the sealing members **100***a*, **100***b*, **100***c* are compressed and deformed against support structure **6**. Because the gaskets **100** and **400** each have a base **100***d* and **404**, respectively, from which the sealing members extend, at least some of the sealing members on each of the gaskets are compressed and deformed against the opposite base **100***d* and **404**, respectively. The inter-engagement of sealing members **100***a*, **100***b*, **100***c* of gasket **100** and sealing members **400***a*, **400***b* and **400***c* of gasket **400** further attenuates the passage of sound. The inter-engagement of sealing members **100***a*, **100***b*, **100***c* of gasket **100** and sealing members **400***a*, **400***b* and **400***c* of gasket **400** and the acoustic barrier **300** cooperate to attenuate the passage of sound through the trailing edge **14** of the door **2**.

The seal for the door **2** may further comprise a channel member **94** that is disposed generally vertically and faces the leading edge **16** of the door **2**. The channel member **94** is dimensioned to receive the edge **16** of the door. The channel member **94** may comprise gaskets **96** and **98** that face both sides of the door such that when the door is in the closed position and the leading edge **16** of the door is received in the channel **94**, gasket **96** engages the outer face of the door **2** near leading edge **16** and gaskets **98** engage the inner face of the door **2** near the leading edge **16** to create the seal. The gaskets **96** and **98** may be formed of any suitable compressible, resilient material such as rubber, polymer or the like. The gaskets **96** and **98** extend the height of the channel member **94** and door **2** to create a seal along the height of the door. In some embodiments each of the gaskets **96** and **98** may comprise at least one elongated sealing member that forms a seal against the door **2**. The gaskets may be formed of any suitable compressible, resilient material such as rubber, polymer or the like. The gaskets **96** and **98** may be supported in elongated slots or holders formed on or mounted to the channel member **94** or by other securement mechanisms.

To seal the bottom of the door **2**, the bottom edge **12** of the door **2** may include a bottom seal that is moved into engagement with the floor when the door is in the closed position as will hereinafter be described. The arrangement of the gaskets as described above and the bottom seal create a door seal that extends about the periphery of the door to prevent the movement of air, sound, light, smoke or the like from moving through the door opening when the door is closed.

Referring to FIGS. **1-9**, the door **2** is supported by and suspended from the trollies **40** where one of the trollies **40** is mounted near the front edge **16** of the door and the other one of the trollies **40** is mounted near the rear edge **14** of the door. The trollies **40** are identical such that one of the trollies is described in detail. The trolley **40** comprises a carriage **42** that rotatably supports a plurality of wheels **44** that are disposed such that the wheels ride on the support surfaces **28** of track **20**. The wheels **44** are arranged in pairs where in the illustrated embodiment, two pairs of wheels are provided. The trolley **40** is mounted on the door **2** using an apron assembly. The carriage **42** includes a threaded hole for receiving a threaded fastener **46** such as a screw such that the carriage **42** may be moved axially along the length of screw when the carriage **42** is rotated relative to the screw. A plate **48** is fixed to the top edge **10** of the door **4** trapping the head of screw between the door and the plate **48**. The plate **48** may be positioned in a pocket **49** formed in the edge of the door **2** such that the plate **48** is flush with the top edge **10**. The plate **48** may be fixed to the top edge **10** of the door **4** by suitable fasteners such as screws that are inserted through holes **47** and threaded into the door. A lock nut **45** may be provided to clamp the plate **48** between the head of the screw **46** and the nut **45** to fix the screw in position. The carriage **42** may be threaded along the length of the screw **46** to adjust the position of the carriage **42** and wheels **44** relative to the door to position the door **2** a set distance below the track and to adjust the position of the door **2** relative to the floor in order to account for variations in the construction of the door **2**. The fastener **46** extends into the gap between the support surfaces **28** such that the wheels **44** are supported on to support surfaces **28**.

An activator pin assembly **60** is mounted in the top edge **10** of the door **4** using a separate apron assembly from that used to mount the trollies **40** to the door. The activator pin assembly **60** comprises an activator pin **62** having a threaded end **62***a* that is threaded into a threaded sleeve **64**. The threaded sleeve **64** is retained in a hole **66** formed in the top edge **10** of the door **4**. A plate **66** may be fixed to the top edge of the door to retain the threaded sleeve **64** between the door and the plate and thereby fix the activator pin **62** to the door **2**. The plate **66** may be positioned in a pocket **68** formed in the edge of the door **4** such that the plate **66** is flush with the surface of the top edge **10**. The plate **66** may be fixed to the top edge of the door **2** by suitable fasteners such as screws that are inserted through holes **68** and threaded into the door. A lock nut **70** and lock washer **72** may be provided to fix the position of the activator pin **62**. Rotation of the activator pin **62** relative to sleeve **64** and lock nut **70** increases or decreases the height of the activator pin **62** relative to the top edge of the door **2** such that the height of the pin **62** is adjustable.

In one embodiment, the activator pin **62** is located in the center of the door **2** approximately midway between the leading edge **16** and the trailing edge **14** and may be positioned proximate to the vertical centerline of the door **2**. The single activator pin **62** may be used to provide soft close functionality in both the opening direction and in the closing direction such that it is not necessary to provide two activators. While the activator pin **62** is described as being positioned approximately midway between the leading edge **16** and the trailing edge **14** it may be slightly offset from the center of the door **2** to allow the door to overlap with the opening as shown in FIG. **3**.

Referring more specifically to FIG. **6**, a pair of soft close devices **50***a*, **50***b* are used in conjunction with the single activator pin **62** such that the single activator pin **62** interacts with both soft close devices **50***a*, **50***b*. Soft close devices are known and a variety of different types of soft close devices may be used to interact with the activator pin **62** to control the opening and closing of the door **2**. A typical soft close device includes a housing **52** that retains a carrier or catch **54** that is engaged in a first position by the activator pin **62** as the door **2** reaches the end of travel. The carrier or catch **54** is operatively connected to a damping device such as spring, pneumatic or gas cylinder or the like, or combinations of such elements, that move the carrier/catch and the door **2** via the engagement of the carrier/catch with the activator pin **62**, from the first position near the fully closed or near the fully open position to a second position at the fully closed or the fully open positions. As the door moves to the fully closed position the activator pin **62** engages the catch/carrier **54** of the first soft close device **50***a* and when the door **2** moves to the fully open position the activator pin **62** engages the carrier/catch **54** of the second soft close device **50***b*. The soft close device **50***a* moves the door **2** from the first position near the fully closed position (FIGS. **2** and **7**) to a second position at the fully closed position (FIGS. **3** and **8**). The soft close device **50***b* operates in the same manner for the fully open position. The damping device moves the door **2** in a controlled manner such that the door **2** does not open or close with too much force. The carrier or catch **54** is moved from the second position to the first position by the reverse movement of the door **2** and/or by a separate biasing mechanism. The soft close devices **50***a*, **50***b* are positioned in the track **20** in the upper interior space **21**. Set screws **23** engage the soft close devices **50***a* and **50***b* to clamp the soft close devices in position against the track **20**.

With a single activator pin **62** approximately centrally positioned on the door **2**, the soft close devices **50** are positioned such that the first soft close device **50***a* is engaged by the activator pin **62** as the door nears the fully closed position and the second soft close device **50***b* is engaged by the activator pin **62** as the door **2** nears the fully open position. When the door **2** is at the halfway point of travel between the fully open position and the fully closed position (FIG. **1**), the soft close devices **50***a*, **50***b* are spaced approximately equally from the activator pin **62** allowing for the overlap of the door **2** against the door frame **4** when the door **2** is in the closed position. In some embodiments, the activator pin **62** may be located offset from the centerline of the door **2**. In such an embodiment, the soft close devices **50***a*, **50***b* may be located in more non-symmetrical positions to maintain the movement of the door **2**. For example, referring to FIG. **1** if the activator pin **62** is moved left of the centerline, the soft close devices **50***a*, **50***b* may be moved to the left the same distance. The distance between the activator pin **62** and the soft close devices **50***a*, **50***b* determines the amount of travel the door **2** moves in the opening and closing directions. The actual spacing between the soft close devices **50***a*, **50***b* depends on the width of the door **2**, the width of the opening and the desired amount of travel of the door **2**.

The use of a single activator pin **62** simplifies the installation of the system, minimizes the number of components and makes alignment of the door **2** easier. It will be appreciated that the use of a single activator pin **62** eliminates the need for a second activator pin. Using a single activator **62** also minimizes door preparation because only a single pin activator mounting structure needs to be installed on the door **2**. Moreover, by locating the single activator pin **62** at the center of the door **2**, the angle of the door **2** and the heights of the edges of the door **2** may be adjusted without moving the activator pin **62** significantly relative to the first and second soft close devices **50***a*, **50***b* making installation of the door **2** easier. Also, using an activator pin **62** that is mounted to the door **2** separate from the trollies **40**, the position of the trollies **40** relative to the door **2** can be adjusted independently of the position of the activator pin **62** relative to the door **2**. In some embodiments, a single soft close device may be used adjacent one of the open and closed positions.

Referring to FIGS. **11** through **16** an integrated guide system and bottom door seal will be described. The door seal may include a bottom door seal **118** that seals the bottom edge **12** of the door **2** against the floor or other support structure such that the door **2** is sealed about its entire periphery when the door **2** is in the closed position. Bottom door seals are known where the door seal is moved vertically from a retracted position (FIG. **14**A) to an extended position (FIG. **14**B) when the door **2** is moved to the closed position such that a sealing member **122** of the bottom door seal **118** engages the floor. One example of such a bottom door seal is disclosed in U.S. Pat. No. 6,125,584 issued to Pemko Manufacturing Co. on Oct. 3, 2000 which is incorporated by reference herein in its entirety. The bottom door seal **118** uses a drive mechanism **120** to move sealing member **122** vertically from the retracted position to the extended position. In one embodiment a plurality of springs **124** are used to move the sealing member **120** vertically between the retracted and extended positions. The springs **124** may be mounted between a plurality of blocks **126** where the ends of each of the springs **124** are mounted to the adjacent blocks and the center of the each of the springs **124** is mounted to the sealing member **122**. At least one of the blocks is slidable laterally relative to the housing and at least one of the blocks is mounted in a fixed position relative to the housing. A push rod **130** (FIG. **16**) is operatively connected to the slide blocks such that when the push rod **130** is moved to actuate the door seal (such as when the door is closed and the push rod engages the vertical wall of channel member **94**), the movement of the push rod **130** is transmitted to the slide blocks via the springs **124** such that the slide blocks are also moved laterally. Because the fixed block is fixed in position, the movement of the slide blocks causes the springs to deform such that the springs force the sealing member **122** vertically downward to seal against a floor or other structure. The sealing member **122** may comprise a deformable gasket **132** that deforms into engagement with the floor to create a tight seal. Although a specific example of a bottom door seal has been shown and described it is to be understood that the integrated guide assembly **200** of the invention may be used with any bottom door seal **118**. The gasket **132** may be formed of any suitable compressible, resilient material such as rubber, polymer or the like such that the gasket deforms into engagement with the floor.

The integrated guide assembly **200** comprises a housing **202** that fits into a groove **204** formed in the bottom edge **12** of the door **2**. The groove **204** extends along and is open to the bottom edge **12** of the door **2**. The groove **204** extends parallel to the direction of travel of the door **2**. Housing **202** fits into the groove **204** such that the housing **202** and the groove **204** extend for the width of the door **2** between the leading and trailing edges **14**, **16** of the door **2**. The housing **202** defines a first channel **208** and a second channel **210**. Both of channels **208**, **210** extend along the length of the housing **202** and for the width of the door **2** and both channels **208**, **210** open toward the bottom edge **12** of the door **2**. In one embodiment a common interior wall **212** defines the interior side of both of the channels **208**, **210** such that the channels **208**, **210** are disposed closely adjacent and parallel to one another to minimize the width of the housing **202**. Channel **208** is further defined by outer wall **214** and channel **210** is defined by outer wall **216**. Channel **208** is closed by top wall **218** that extends between the interior wall **212** and the outer wall **214** and channel **210** is closed by top wall **220** that extends between the interior wall **212** and the outer wall **216**. In the illustrated embodiment the housing **202** includes a pair of opposed flanges **222**, **224** that extend from outer walls **214** and **216**, respectively and that may be secured to the bottom edge **12** of the door **2**. The flanges may be eliminated such that the housing **202** terminates at the end of or inside of the groove **204**. The housing **202** may comprise a unitary one-piece member such that the channel **210** for receiving the guide member **232** and the channel **208** for receiving the bottom seal **118** are part of the same member. For example, in some embodiments, the housing **202** may be an extruded member. As used herein the housing **202** may be considered a unitary one-piece member where the housing **202**, including both channels **208** and **210**, is made of separate components provided the components are connected together outside of the door **2** and are installed in the door **2** as a unit. For example, the housing **202** may be made of planar members secured to one another by welding, crimping, fasteners or the like.

The first channel **208** is dimensioned to receive the bottom seal **118** such that the sealing member **122** may be retracted and extended to seal with the floor or other surface as shown in FIGS. **14**A and **14**B. The second channel **210** is dimensioned and configured to slidably receive the door guide **230**. The door guide **230** comprises an upright guide member **232** that, when engaged with channel **210** maintains the bottom edge **12** of the door **2** in position relative to the wall as the door **2** is opened and closed. In the illustrated embodiment the guide member **232** comprises a flat plate that is closely but slidably received in channel **210**. The door guide **230** further comprises a support member **240** that supports the guide member. The support member **240** is arranged such that it does not interfere with the engagement of the sealing member **122** with the floor when the sealing member **122** is extended from the housing.

The support member **240** is positioned such that it extends away from and behind the sealing member **122**. In one embodiment the support member **240** comprises an L-shaped member having a horizontal section or leg **242** that may extend along the floor and may be secured thereto and a vertical portion or leg **244** that may be secured to the wall behind the door. It is important for the sealing member **122** to be able to make contact with the floor along its entire length in order to provide a suitable barrier against sound, air, light etc. Therefore, arranging the support **240** such that it does not extend under the sealing member **122** allows the seal to contact the floor in uninterrupted manner. The guide member **232** has a front side **232***a* that faces toward the bottom seal **118** and a back side **232***b* that faces away from the bottom seal **118**. In one embodiment, where the channels are disposed closely adjacent to one another the support member **240** does not extend beyond the front side **232***a* such that it does not extend under the sealing member **122**.

End plates **248**, **250** may be mounted to the side edges of the door **2** to cover the bottom seal. End plate **248** includes a slot **252** to allow the guide member **232** to extend through the end plate **248**. Endplate **250** includes an aperture **254** that allows the push rod **130** to extend through end plate **250**. In one embodiment screws **260** may extend through apertures in the end plates and engage threaded holes **262** in the housing **202** to secure the end plates **248**, **250** in position.

The integrated guide system and bottom door seal of FIGS. **11**-**16** may be used with the peripheral door seal and soft close system described with respect to FIGS. **1**-**10**. However, the integrated guide system and bottom door seal, the peripheral door seal and the soft close system may be used independently of one another. When used together the integrated guide system and bottom door seal, the peripheral door seal provide a sealing system that resists the transmission of air, light, sound across the door.

Although specific embodiments have been illustrated and described herein, those of ordinary skill in the art appreciate that any arrangement which is calculated to achieve the same purpose may be substituted for the specific embodiments shown and that the invention has other applications in other environments. This application is intended to cover any adaptations or variations of the present invention. The following claims are in no way intended to limit the scope of the invention to the specific embodiments described herein.

The invention claimed is:

1. A sliding door system comprising:

a door comprising a top edge, a leading edge and a trailing edge defining an inner face, the door being mounted for sliding movement relative to a door opening formed in a structure between a closed position and an open position;

a first gasket and a second gasket deformed into engagement with the inner face adjacent the top edge, the first gasket being spaced from the second gasket to create a gap between the first gasket, the second gasket and the inner face; and

an acoustic barrier comprising a front side having a pad and a rear side having a plate positioned at one end of the first gasket and the second gasket, wherein at least a portion of the plate is dimensioned to extend between the first gasket and the second gasket, and wherein the pad of the acoustic barrier engages the inner face of the door and closes the gap between the first gasket, the second gasket and the inner face when the door is in the closed position.

2. The sliding door system of claim **1** wherein the front side of the acoustic barrier is generally planar and deformable.

3. The sliding door system of claim **1** wherein the plate supports the pad.

4. The sliding door system of claim **1** wherein the pad comprises a pile pad.

5. The sliding door system of claim **1** wherein a back side of the acoustic barrier comprise slots that receive flanges of a bracket for aligning with the first gasket and the second gasket.

6. The sliding door system of claim **1** wherein the acoustic barrier is fixed to the structure.

7. The sliding door system of claim **1** further comprising a third gasket mounted to the structure such that it is positioned adjacent the trailing edge of the door when the door is in the closed position and a fourth gasket mounted to the trailing edge of the door.

8. The sliding door system of claim **7** wherein the third gasket comprises a plurality of first deformable sealing members and the fourth gasket comprises a plurality of second deformable sealing members.

9. The sliding door system of claim **8** wherein the plurality of first deformable sealing members interdigitate with the plurality of second deformable sealing members when the door is in the closed position.

10. The sliding door system of claim **8** wherein the plurality of first deformable sealing members alternate with the plurality of second deformable sealing members from the leading edge of the door toward the trailing edge of the door when the door is in the closed position.

11. The sliding door system of claim **8** wherein the third gasket is mounted to the structure such that the plurality of first deformable sealing members extend at a non-perpendicular angle relative to the door.

12. The sliding door system of claim **7** further comprising a channel member that is disposed generally vertically and receives the leading edge of the door when the door is in the closed position, the channel member comprising a fifth gasket that faces the door such that when the door is in the closed position and the leading edge of the door is received in the channel member such that the fifth gasket engages the door near the leading edge.

13. A sliding door system comprising:

a door comprising a top edge, a leading edge and a trailing edge defining an inner face, the door being mounted for sliding movement relative to a door opening formed in a structure between a closed position and an open position;

a first gasket and a second gasket deformed into engagement with the inner face adjacent the top edge, the first gasket being spaced from the second gasket to create a gap between the first gasket, the second gasket and the inner face; and

an acoustic barrier positioned at one end of the first gasket and the second gasket, wherein a back side of the acoustic barrier comprises slots that receive flanges of a bracket for aligning with the first gasket and the second gasket, and wherein the acoustic barrier engages the inner face and closes the gap between the first gasket, the second gasket and the inner face when the door is in the closed position.

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